

La Fine

Stefano Mizzaro

DMIF – UNIUD

mizzaro@uniud.it

Social Computing – Lezione 24

9 gennaio 2026

Scaletta

- Fine accordo
- Fake news
- Riassunto corso
- Esame
 - Regole
 - Esempi di domande

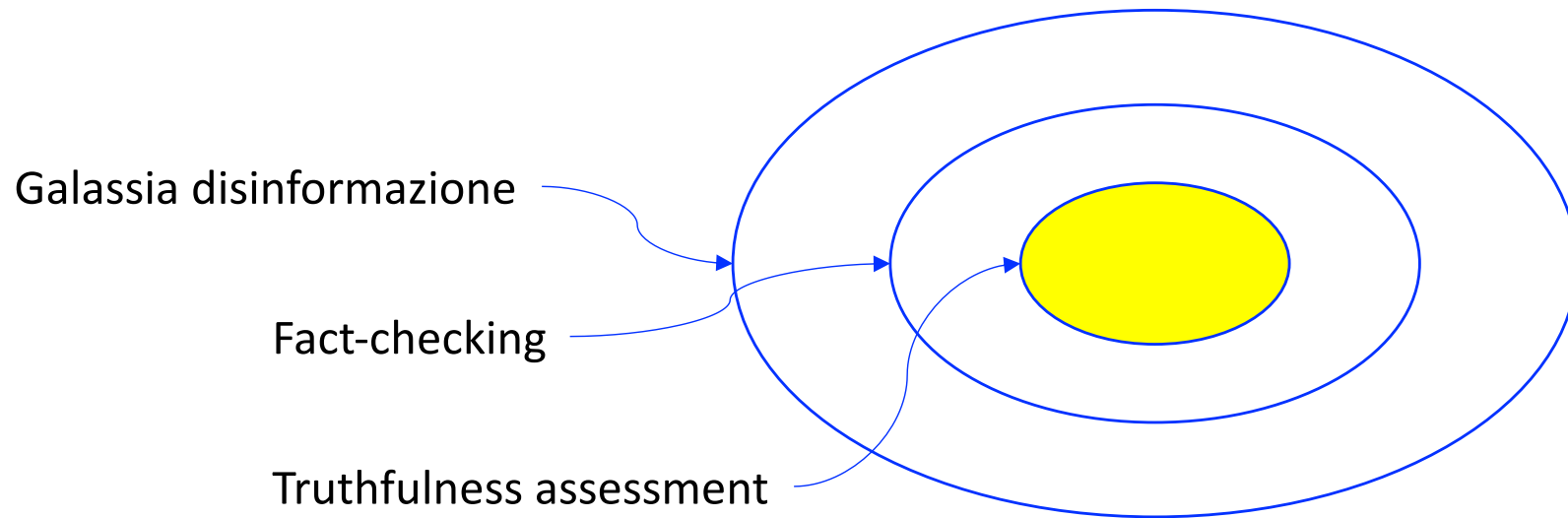
Scaletta

- Fine accordo
- **Fake news**
- Riassunto corso
- Esame
 - Regole
 - Esempi di domande

Back to "Fake News"

- Disinformazione, Infodemia, ...
- Progetto / domanda generale:
 - Combattere la disinformazione con il Crowdsourcing?
 - Si può usare il crowdsourcing per identificare le fake news?
 - Il crowd è in grado di distinguere notizie vere e false?
 - Il crowd è in grado di dire se una notizia è vera o no?
- Sembra difficile...
- Ci stiamo lavorando da qualche anno
- Aspetti specifici:
 - Influenza del "bias" politico (...)
 - Effetti della scala usata
 - Ci siamo anche concentrati sulle news relative al COVID-19
 - Longitudinale, Variazione nel tempo
 - Integrazione con AI
 - ...

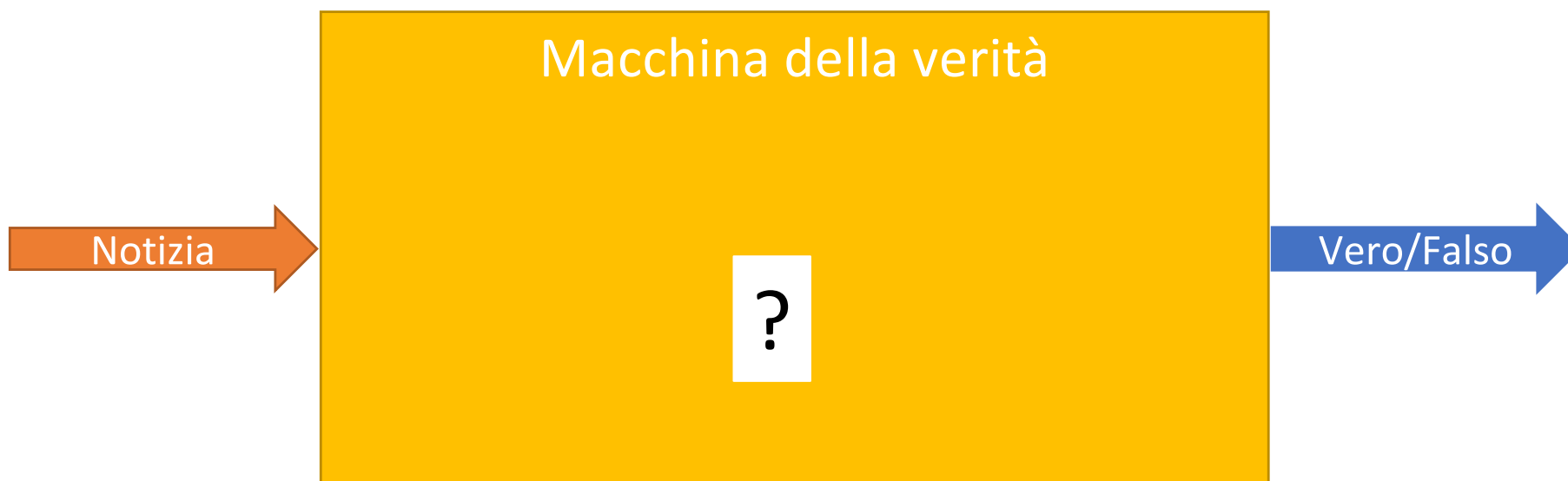
"Poco e bene"



«lo stimo più il trovar un vero, benché di cosa leggiera, che 'l disputar lungamente delle massime questioni senza conseguir verità nissuna»

(Galileo Galilei, anche se non c'è evidenza nei suoi scritti)

"Macchina della verità", "Truthfulness assessment machine"



- Come costruirla?
- Cosa metterci lì?

Esperti?



- Eh non ci sono abbastanza esperti!
- (o, c'è troppa disinformazione)

AI?



- Classificazione (es.: filtro anti spam)
- Prestazioni super-umane
- Oggi: LLM (Large Language Model)

Crowdsourcing?



- Potrebbe essere una buona "via di mezzo" fra esperti e AI

AI + Crowdsourcing + Esperti?!

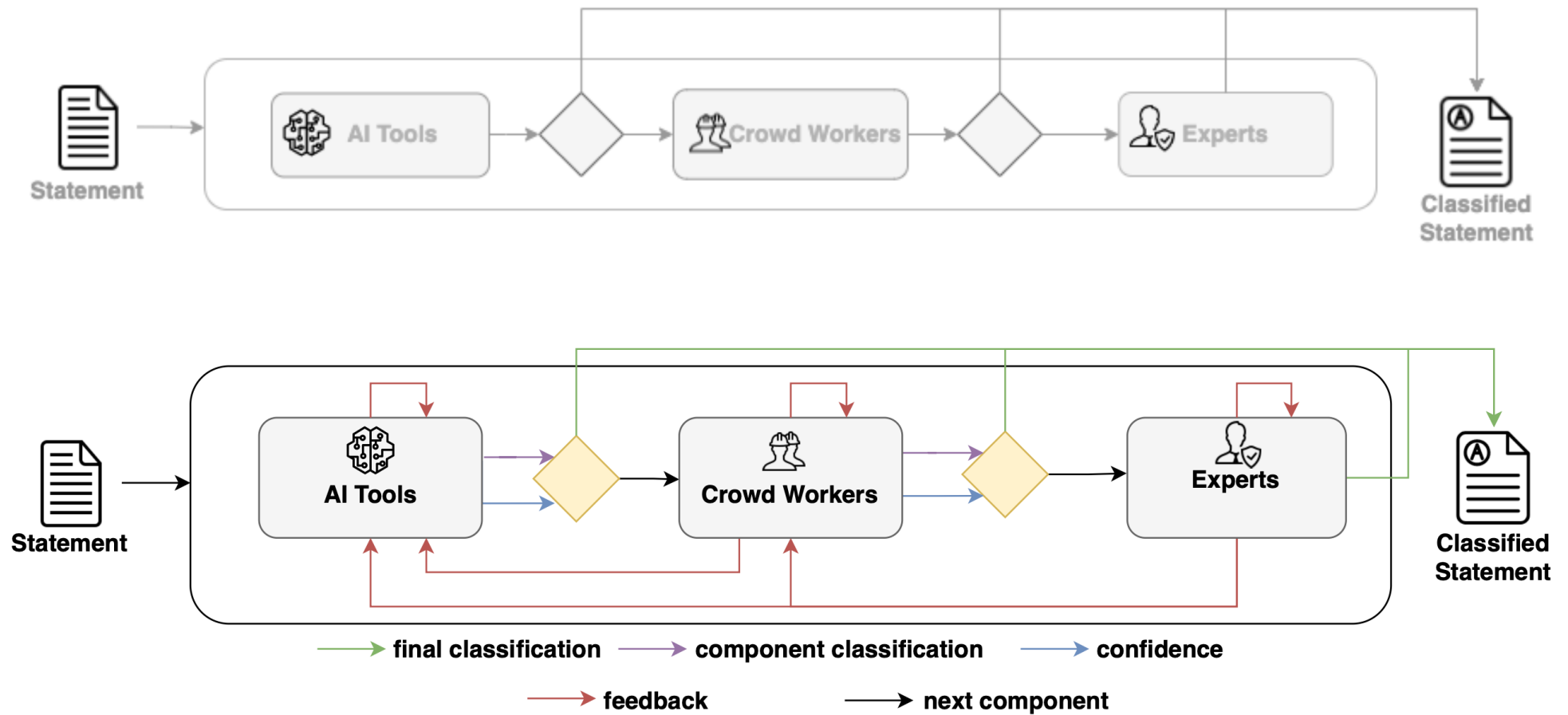


- E perché non tutti insieme?

AI + Crowdsourcing + Esperti

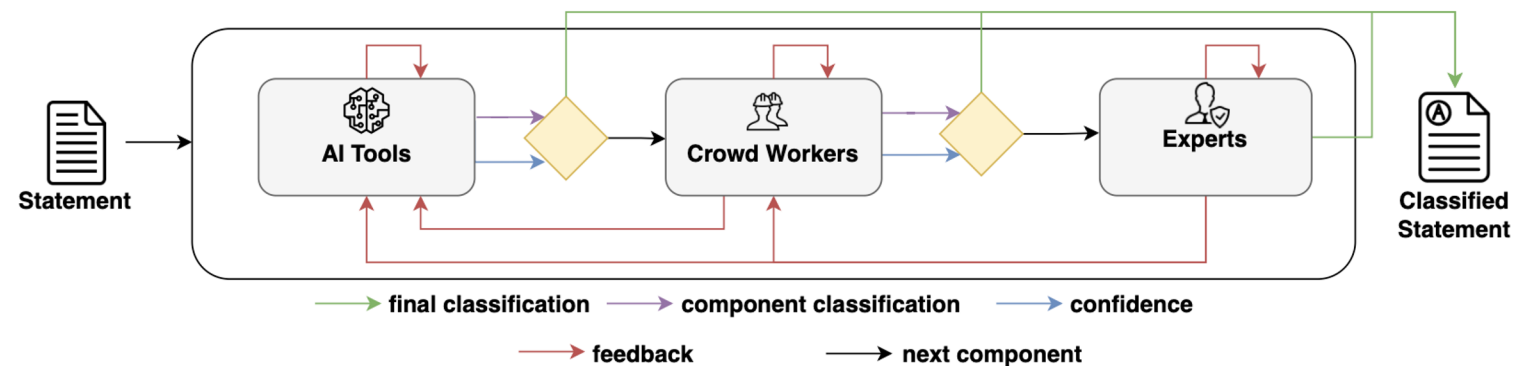


AI + Crowdsourcing + Esperti



Vari temi specifici

- Ha senso?
- Efficacia delle singole componenti
- Confidenza?
- Scala dell'output
- Valutazione
- Bias/Pregiudizi
- AI: che prompt?
- Evidenza
- Deepfake
- ...



Ha senso?

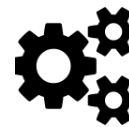
- Il crowd è in grado di dire se una notizia è falsa o vera?
- Perché potrebbe non avere senso



"Napalm51"

Ha senso?

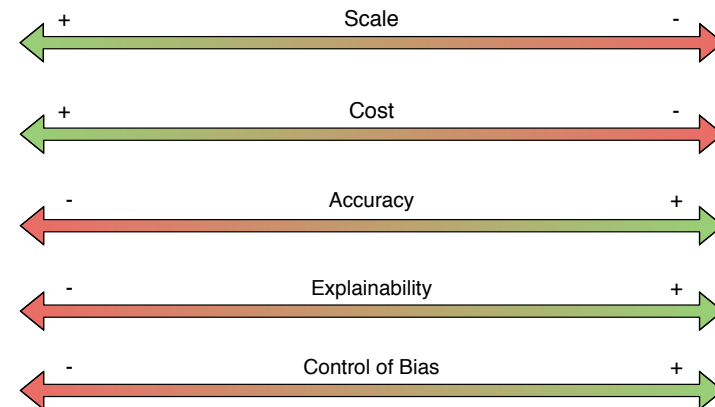
- Perché potrebbe avere senso
 - AI vs. Esperti



AI Tools

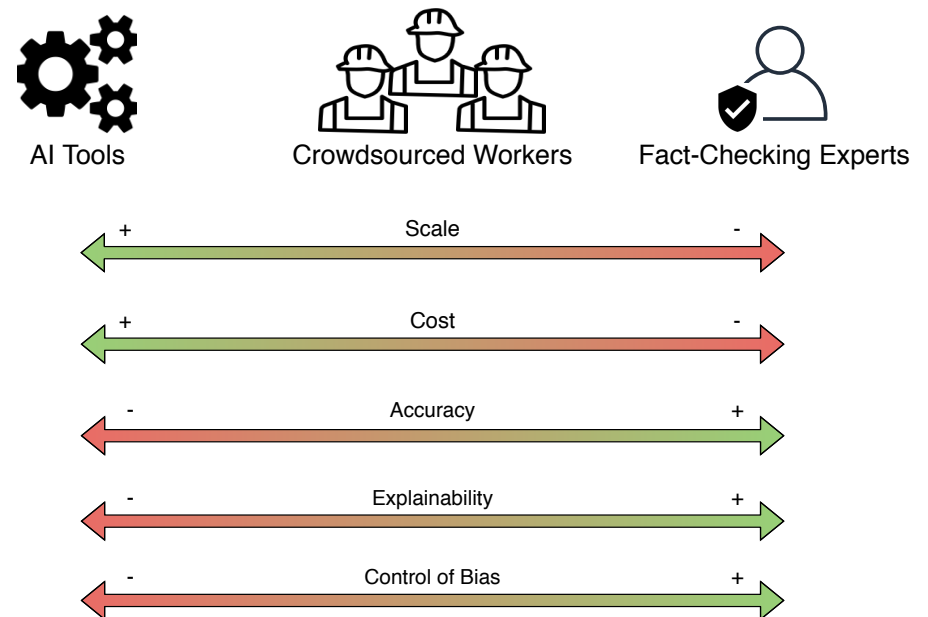


Fact-Checking Experts



Ha senso?

- Perché potrebbe avere senso
 - AI vs. Esperti
 - E vs. Crowdsourcing
- Tentativi anche IRL
 - Community notes

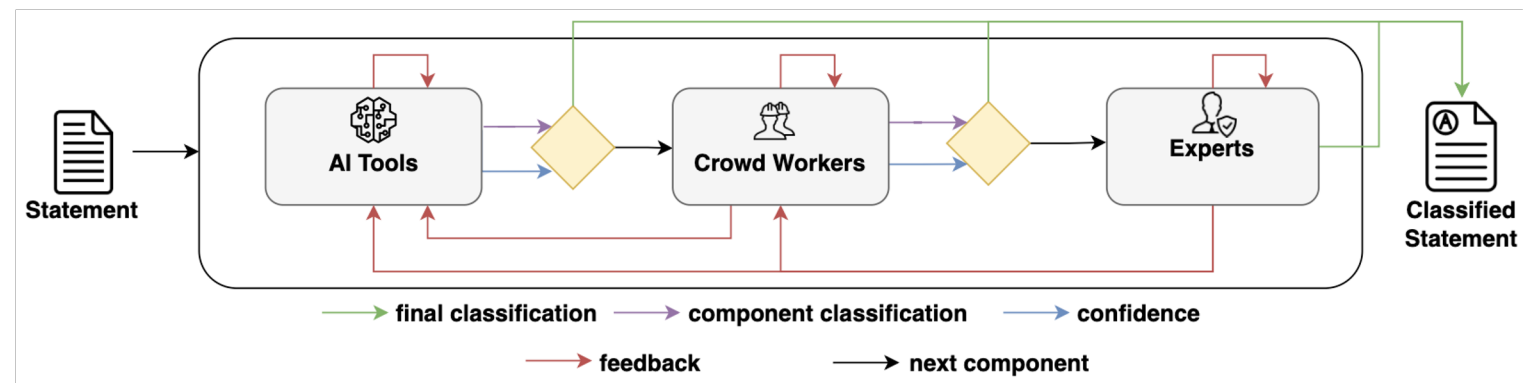


Community Notes

- Was Birdwatch
- Su X (Twitter)
 - Ma anche su Facebook, Instagram, ...
- La comunità (utenti X) aggiunge note ai post
 - Ad es. per segnalare fake news (ma non solo)
- Le note vengono "votate"
- Algoritmo di aggregazione
 - Le note con voti migliori vengono mostrate
 - Aggregazione non a maggioranza ma richiede accordo fra utenti che di solito sono in disaccordo

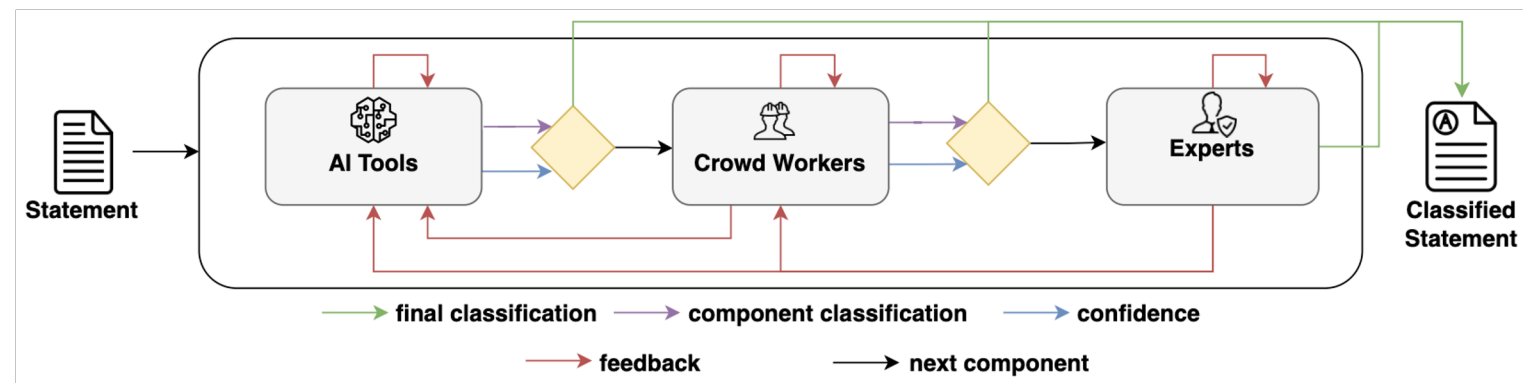
Efficacia delle singole componenti

- Esperti: 100% ca.
- Crowd: 80% ca.
- AI: 80% ca.
- Nota: potrebbe bastare $< 100\%$...
 - ... visto il volume di disinformazione (es.: filtro antispam; Google)
 - "La perfezione non è di questo mondo"
- Ma serve Crowd $>$ AI, altrimenti lo si può eliminare!



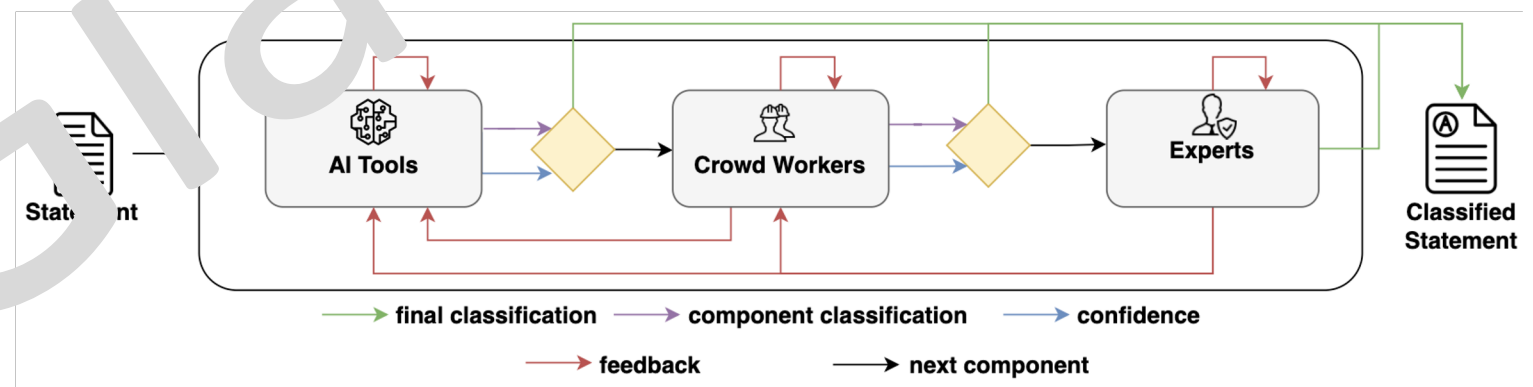
Confidenza

- Quando scegliere di dare l'output e quando passare al prossimo?
- Confidenza (fiducia) nella propria risposta
- Le confidenze delle AI non sono affidabili (per ora?)
- "Unknoww unknowns"
 - Affermazioni giudicate in modo scorretto ma con convinzione



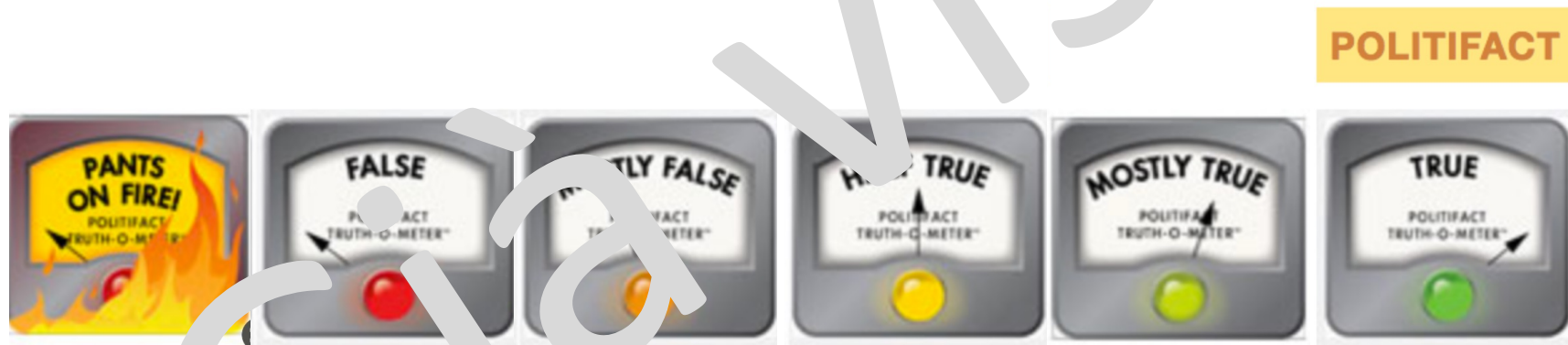
Scala dell'output

- Falso, Vero
- Falso, Né falso né vero, Vero
- Falsissimo, Falso, Vero, Verissimo
- Completamente falso, parzialmente falso, parzialmente vero, completamente vero
- Altro?



6 livelli, e con quelle etichette...

- Politifact: <https://www.politifact.com>
- IRL, non solo curiosità accademica



POLITIFACT

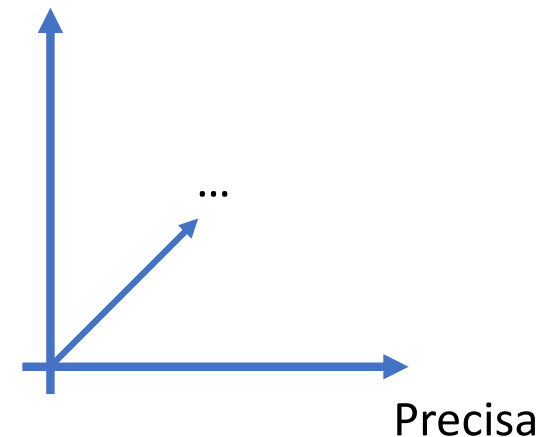
The Poynter Institute

Ma anche

- 100 livelli
- Magnitude estimation (stima della magnitudine)
 - Un numero qualsiasi da zero (escluso) in su
- Preferenze
 - Coppie (A più vero di B)
 - Elenco (ordinare)

O anche peggio

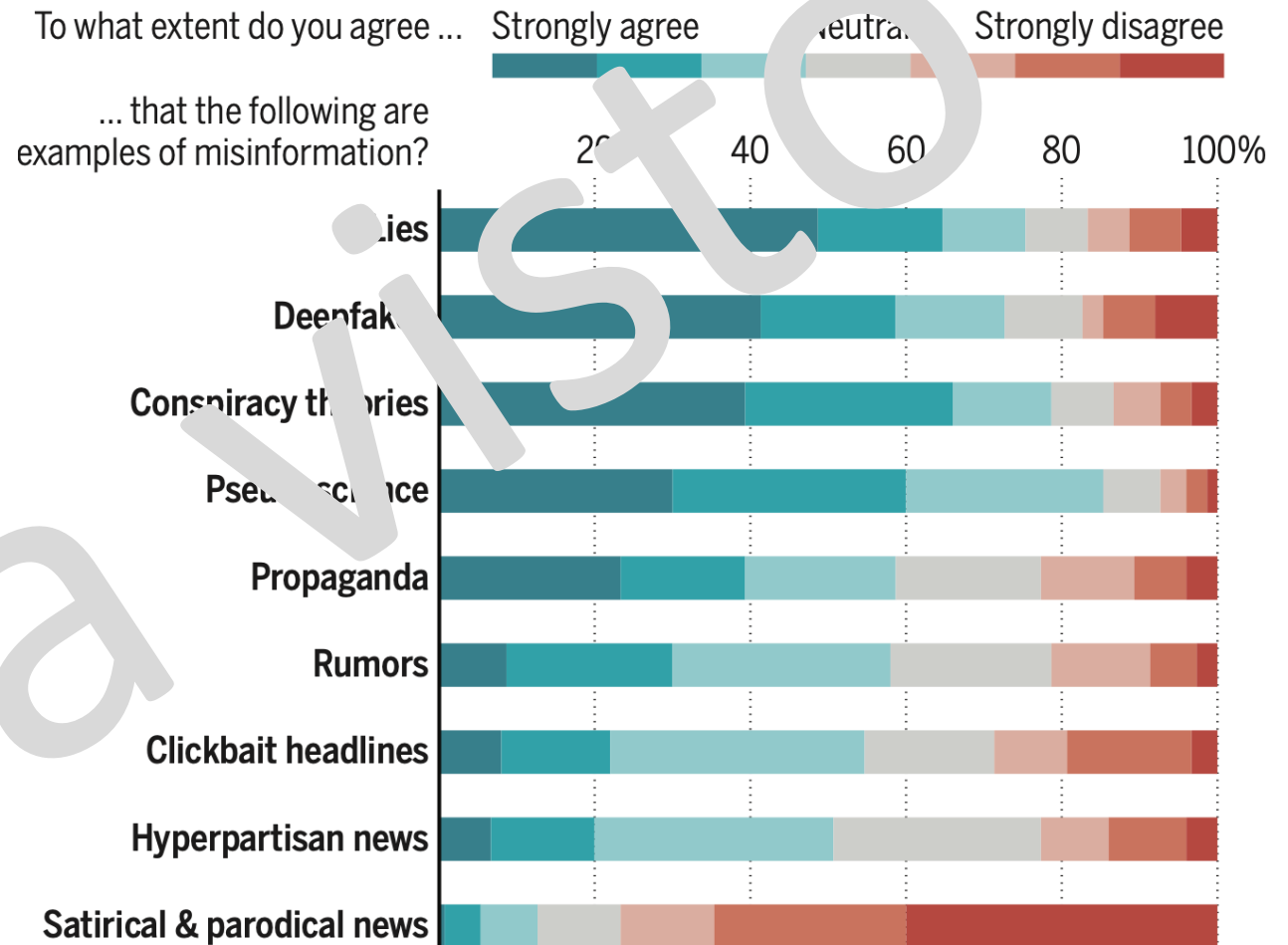
- Unidimensionale → Multidimensionale
 - Per essere vera una notizia deve essere
 - Precisa
 - Completa
 - Neutrale
 - Corretta
 - Comprensibile
 - Ecc. Ecc.
- Varie dimensioni/categorie



Sondaggio

Defining misinformation

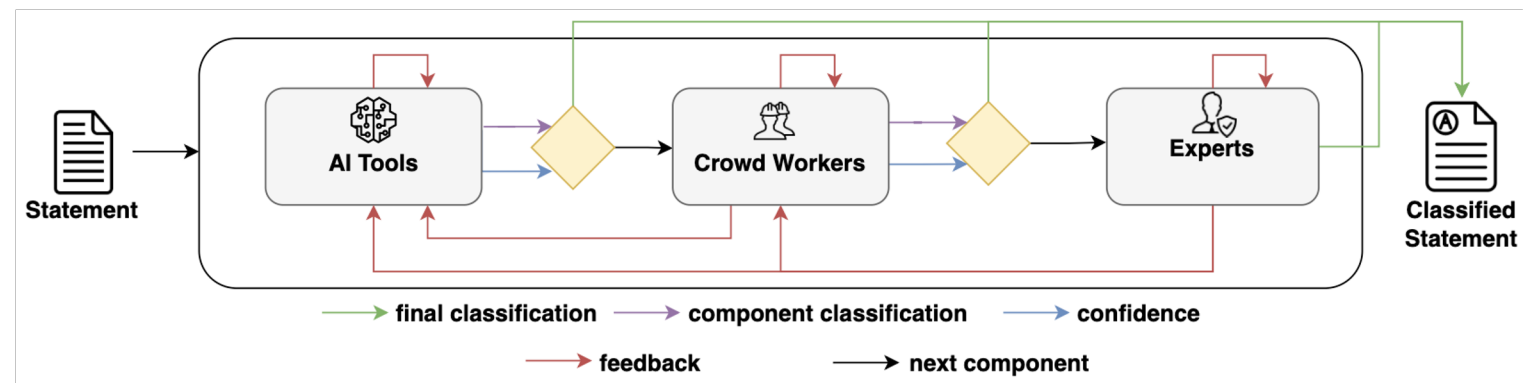
A 2023 survey of 150 misinformation experts found there is often no consensus about which kinds of content count as misinformation.



<https://www.science.org/doi/epdf/10.1126/science.adu2100>

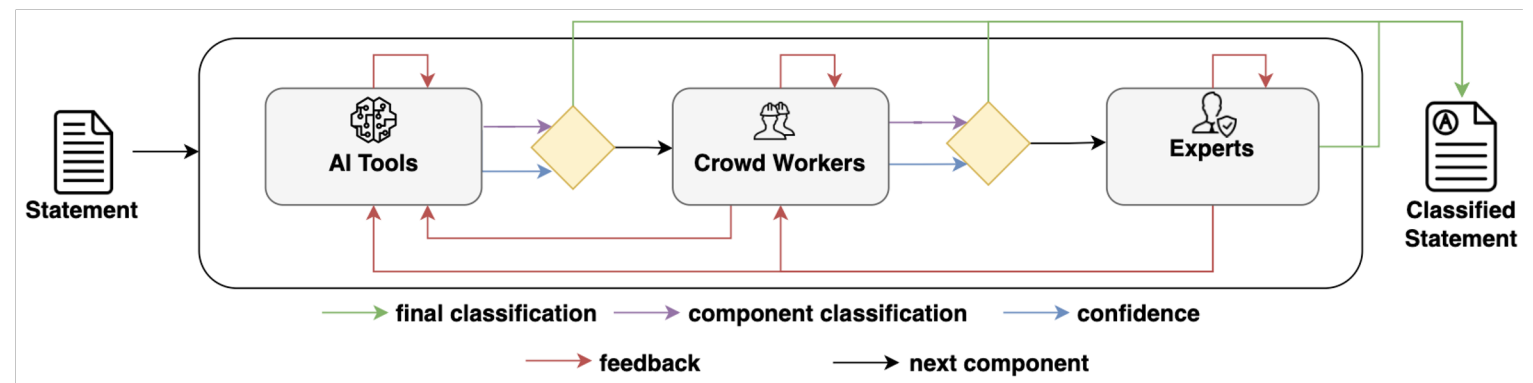
Valutazione

- 2 versioni della Macchina della verità, quale va meglio?
- «Misura ciò che è misurabile e rendi misurabile ciò che non lo è»
 - (Sempre Galileo, sempre senza evidenza, ma... mi piace)
- Come rispondere?
 - Metodologia



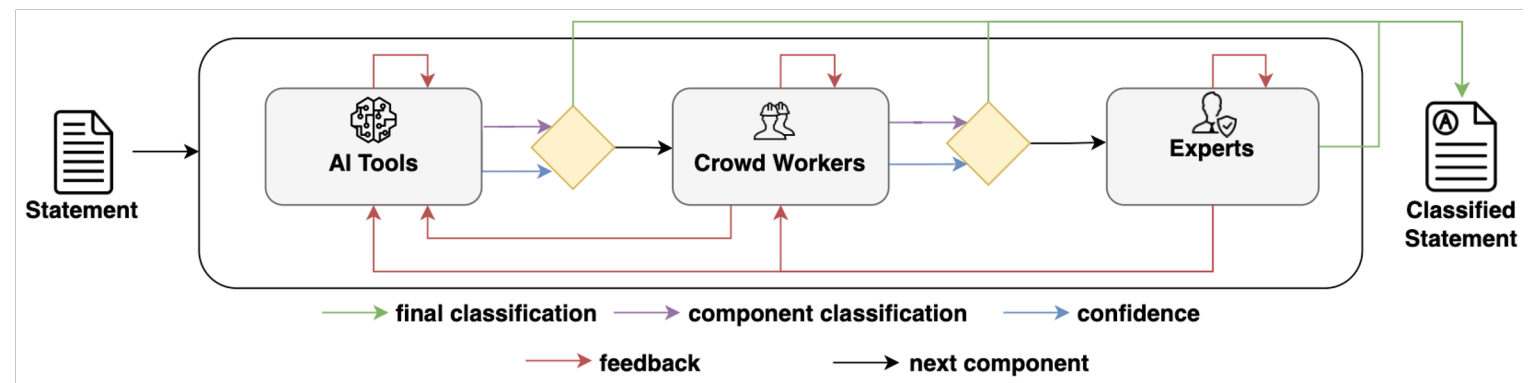
Valutazione

- Benchmark?
 - Data contamination
 - Multidimensionale? Mancanza di dati di training e test



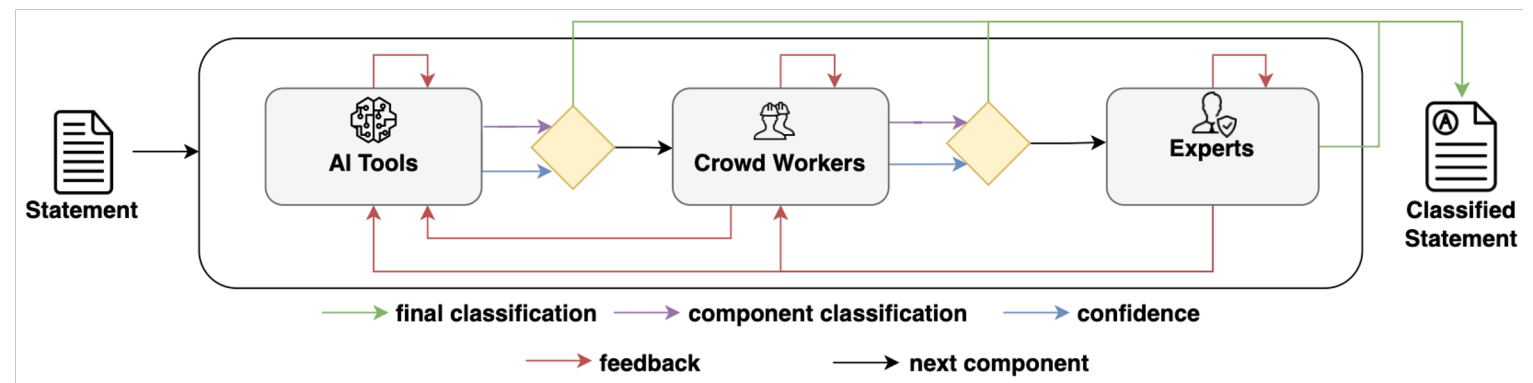
Valutazione

- Scala $\leftrightarrow$ Misura di valutazione
- Accuratezza: OK per scala a 2 valori, non per 3+
- Problema scientifico: scala ordinale, non ci sono misure standard
- (e non chiedetemi come fare per veridicità multidimensionale!)



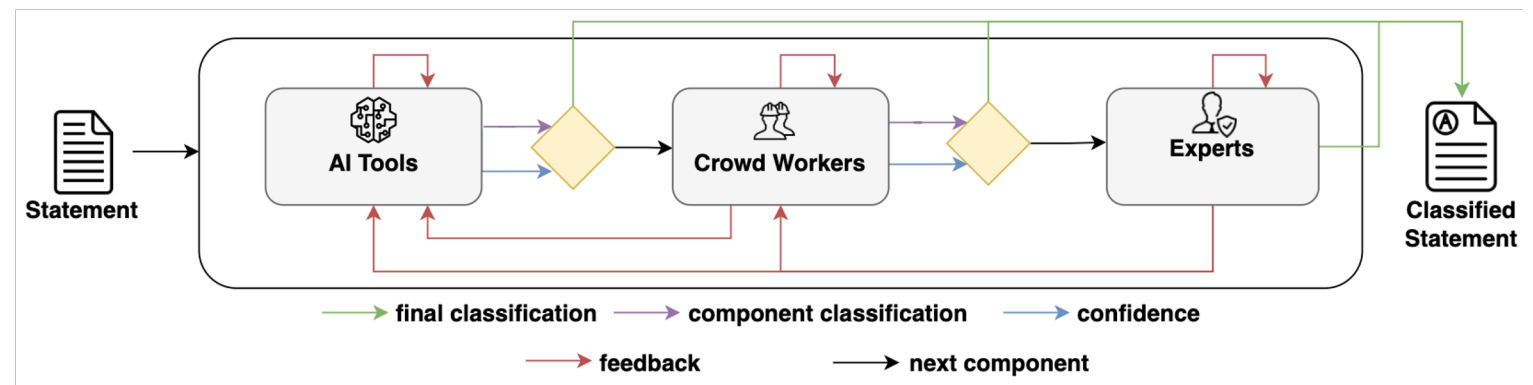
Bias/Pregiudizi

- Gli umani (Esperti e Worker) hanno i loro bias
- Partigianeria politica: giudico più vere le affermazioni fatte da politici della mia parte
 - Pregiudizio di conferma (Confirmation bias)
- E le AI?
 - Eh... (ci si sta lavorando)
 - Rischio censura!



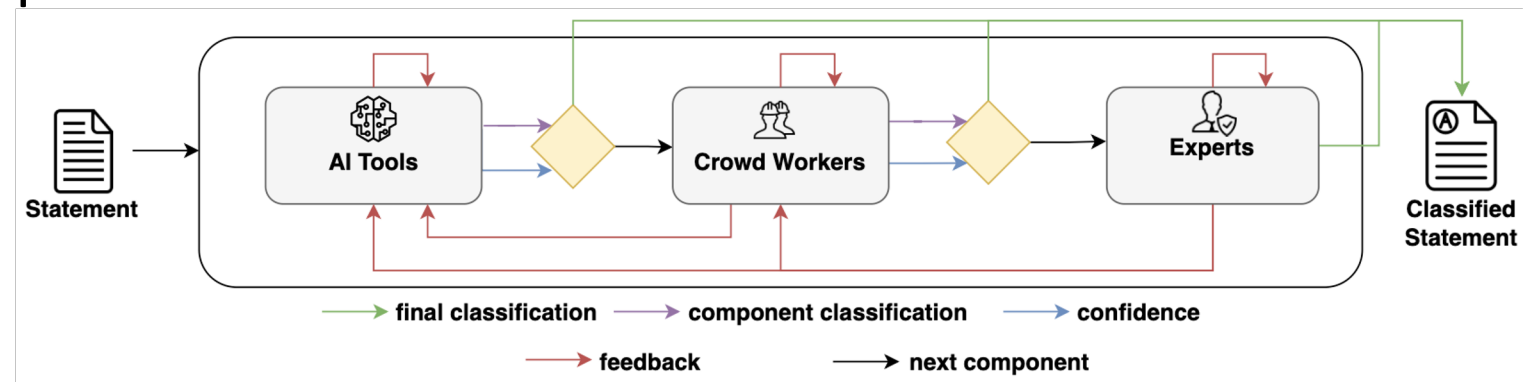
AI: Che prompt?

- Tanti modi di chiedere all'AI "Questa notizia è vera?"
- Per es., includere degli esempi?
- "Evidenza"?



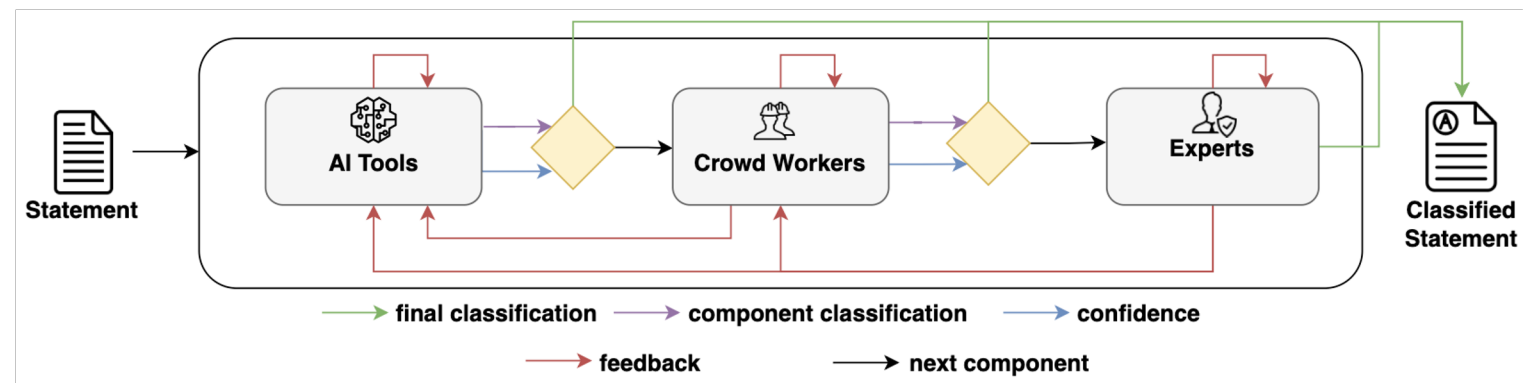
Evidenza

- Come trovarla?
- RAG = Retrieval Augmented Generation
 - La risposta dell'AI è generata dopo un processo di reperimento
 - "AI che usa un motore di ricerca per trovare evidenza che supporti o refuti la notizia"



Deepfake

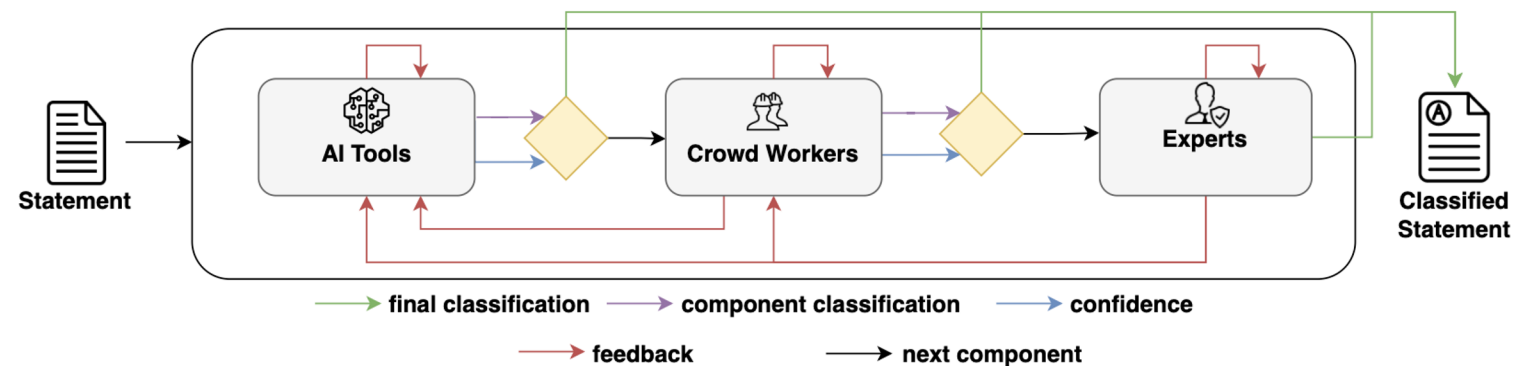
- Video vero o falso?
 - Compito molto difficile
- Neanche il crowd ci riesce...
 - E aggregare dati pessimi peggiora la situazione
 - (la Congettura del Crowdsourcing non vale!)



Vari temi specifici

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Che fatica 😓



"Fake news" e Crowdsourcing

- Vari esperimenti simili
- Abbiamo preso statement già giudicati da esperti (es.: Politifact) e li abbiamo fatti giudicare ai worker

Un po' di videate da un esperimento

TASK INSTRUCTIONS

NO RETAKES

Instructions

In this task, you will be asked to select a value to indicate the **truthfulness level** of some statements.

We will show you **8 statements** made by popular people (for example, political figures) together with the information of who made the statement and in which date. For each statement, we ask you to tell us its "truth level", that is, **how much the statement represents the reality**; how reliable it is. Note that the statement has really been made by the speaker, we are not discussing that: it is the **truthfulness of the content** of the statement that has to be judged. The truth level must be selected among 6 different labels. For example, if the statement totally represents reality you should select "True", while if it is false you should select "Lie".

Note that there are some checks throughout the task, and if you do not perform these correctly you will not be able to terminate and get paid. The data from this task is being gathered for research purposes. No personally-identifying information is recorded. Participation is entirely voluntary, and you are free to discontinue at any point.

Task Description

First, you will be asked to fill in four questionnaires that will help us collect more reliable data.

You will then be asked to rate 8 statements, each of them expressed by a person in a given context. You will be shown the 8 statements, one at a time. For each statement, your task is **to judge the truthfulness and validity of the statement** using the radio buttons. If you wish to change a previously given judgment, you can use the back button to revisit it.

For each judgment, you will also be asked to **provide a URL and a short text to justify your selection** by using our search engine. You will have to choose one of the URLs provided as result to your search queries by such search engine. Finally, you will have to **fill a textarea with your justification**.

The statements are not presented in any particular order. You might see many good statements, many bad ones, or any combination. Try not to anticipate, and simply rate each statement after reading it. Don't try to guess or randomly click, as there are some cross-checks, as indicated above. **Be sure you have filled each field**, otherwise you will not be able to finish the task.

CLOSE

- 1 Questionna... 2 Questionna... 3 Questionna... 4 Questionna... 5 Docume... 6 Docume... 7 Docume... 8 Docume... 9 Docume... 10 Docume... 11 Docume... 12 Docume... 13 Outco...

Please answer each question

1) What is your age range?

- ☐ 0-18
☒ 19-25
☐ 26-35
☐ 36-50
☐ 50-80
☐ 80

2) What is the highest level of school you have completed or the highest degree you have received?

- ☐ High school incomplete or less
☒ High school graduate or GED (includes technical/vocational training that doesn't towards college credit)
☐ Some college (some community college, associate's degree)
☐ Four year college degree/bachelor's degree
☐ Some postgraduate or professional schooling, no postgraduate degree
☐ Postgraduate or professional degree, including master's, doctorate, medical or law degree

3) Last year what was your total family income from all sources, before taxes?

- ☐ Less than 10,000
☒ 10,000 to less than 20,000
☐ 20,000 to less than 30,000
☐ 30,000 to less than 40,000
☐ 40,000 to less than 50,000
☐ 50,000 to less than 75,000
☐ 75,000 to less than 100,000
☐ 100,000 to less than 150,000
☐ 150,000 or more

4) In general, would you describe your political views as

- ☐ Very conservative
☐ Conservative

- ☐ Some postgraduate or professional schooling, no postgraduate degree
- ☐ Postgraduate or professional degree, including master's, doctorate, medical or law degree

3) Last year what was your total family income from all sources, before taxes?

- ☐ Less than 10,000
- ☒ 10,000 to less than 20,000
- ☐ 20,000 to less than 30,000
- ☐ 30,000 to less than 40,000
- ☐ 40,000 to less than 50,000
- ☐ 50,000 to less than 75,000
- ☐ 75,000 to less than 100,000
- ☐ 100,000 to less than 150,000
- ☐ 150,000 or more

4) In general, would you describe your political views as

- ☐ Very conservative
- ☐ Conservative
- ☐ Moderate
- ☐ Liberal
- ☒ Very liberal

5) In politics today, do you consider yourself a

- ☐ Republican
- ☐ Democrat
- ☒ Independent
- ☐ Something else

6) Should the U.S. build a wall along the southern border?

- ☐ Agree
- ☒ Disagree
- ☐ No opinion either way

7) Should the government increase environmental regulations to prevent climate change?

- ☒ Agree
- ☐ Disagree
- ☐ No opinion either way

NEXT

(*) you have to fill each field to proceed



Questionnai...

2

Questionnai...

3

Questionnai...

4

Questionnai...

5

Docume...

6

Docume...

7

Docume...

8

Docume...

9

Docume...

10

Docume...

11

Docume...

12

Docume...

13

Outco...

Please answer each question

1) If three farmers can plant three trees in three hours, how long would it take nine farmers to plant nine trees?

3



NEXT

(*) you have to fill each field to proceed



Questionnai...



Questionnai...

3

Questionnai...

4

Questionnai...

5

Docume...

6

Docume...

7

Docume...

8

Docume...

9

Docume...

10

Docume...

11

Docume...

12

Docume...

13

Outco...

Please answer each question

1) Sean received both the 5th highest and the 5th lowest mark in the class. How many students are there in the class?

9

**NEXT**

(*) you have to fill each field to proceed



Questionnai...



Questionnai...



Questionnai...



Questionnai...



Docume...



Docume...



Docume...



Docume...



Docume...



Docume...



Docume...



Docume...



Outco...

Please answer each question

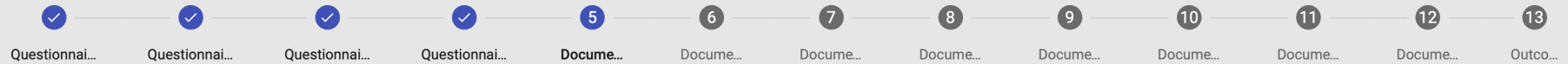
1) In an athletics team, females are four times more likely to win a medal than males. This year the team has won 20 medals so far. How many of these have been won by males?

4



NEXT

(*) you have to fill each field to proceed



Please judge the truthfulness of the following statement

Claimant/Source: Facebook Post

Date: 2020-02-04

Statement: On February 7, the WHO warned about the limited stock of PPE. That same day, the Trump administration announced it was sending 18 tons of masks, gowns and respirators to China.

Please select one of the URLs below to justify the truthfulness value of the statement, then input the truth level as one from the six labels provided and write your justification.

Insert your query

SEARCH

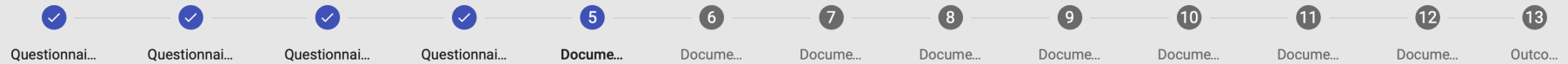
0 results found.

- ☐ Lie
- ☐ False
- ☐ Barely True
- ☐ Half True
- ☐ Mostly True
- ☐ True

Your justification here

NEXT

(*) you have to fill each field to proceed



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February 7 WHO limited stock of PPE trump

SEARCH

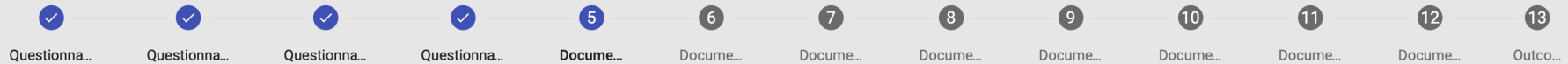
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Insert your query

February 7 WHO limited stock of PPE trump

SEARCH

46 results found.

[Did the Trump Administration Send 18 Tons of PPE to China ...](https://www.snopes.com/fact-check/us-tons-ppe-china/)

<https://www.snopes.com/fact-check/us-tons-ppe-china/>

Did the Trump Administration Send 18 Tons of PPE to China in Early 2020? ... of PPE are not new. Back on February 7, the World Health Organization sounded alarm bells about "the limited stock of ...

Select ☐

[Trump administration sent protective medical gear to China ...](https://www.msn.com/en-us/health/medical/trump-administration-sent-protective-medical-gear-to-china-while-he-was-minimizing-the-virus-threat-to-us/ar-BB11SSuo)

<https://www.msn.com/en-us/health/medical/trump-administration-sent-protective-medical-gear-to-china-while-he-was-minimizing-the-virus-threat-to-us/ar-BB11SSuo>

Concerns about a dwindling supply of PPE are not new. Back on February 7, the World Health Organization sounded alarm bells about "the limited stock of PPE," noting demand was 100 times higher ...

Select ☐

[Trump admin sent medical gear to China while he minimized ...](https://dnyuz.com/2020/03/30/trump-admin-sent-medical-gear-to-china-while-he-minimized-the-virus-threat-to-us/)

<https://dnyuz.com/2020/03/30/trump-admin-sent-medical-gear-to-china-while-he-minimized-the-virus-threat-to-us/>

Concerns about a dwindling supply of PPE are not new. Back on February 7, the World Health Organization sounded alarm bells about "the limited stock of PPE," noting demand was 100 times higher than normal for this equipment.. Yet the same day as the WHO warning, the Trump administration announced that it was transporting to China nearly 17.8 tons (more than 35,000 pounds) of "masks ...

Select ☐

[America Has PPE Shortage Because Trump Sent Ours to China ...](https://www.themarysue.com/shortage-of-ppe-because-trump-sent-our-stockpile-to-china/)

<https://www.themarysue.com/shortage-of-ppe-because-trump-sent-our-stockpile-to-china/>

America Has a Shortage of PPE Because Trump Sent Our Stockpile to China Among all of Trump's poor decisions, this might be the deadliest.

Select ☐

[Reports that Trump sent PPE equipment to China Feb 4 ...](https://2peasrefugees.boards.net/thread/104967/reports-trump-sent-equipment-china)

<https://2peasrefugees.boards.net/thread/104967/reports-trump-sent-equipment-china>

There are reports that Trump sent 17.8 tons of PPE equipment to China on February 4. (tweet from Pompeo @secretary Pompeo February 7. There are also reports that Ottawa

Select ☐

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Fact Checks · Politics

Did the Trump Administration Send 18 Tons of PPE to China in Early 2020?

Critics have questioned the wisdom of exporting to China medical supplies that would soon be vitally needed in the U.S.

DAVID MIKKELSON

PUBLISHED 31 MARCH 2020



Image via Getty Images



AD. ARTICLE CONTINUES BELOW

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Does Trump Benefit Financially by Promoting Hydroxychloroquine as COVID-19 Treatment?

The U.S. president is invested in several mutual funds — some of which include very minor stakes in pharmaceutical companies.

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Claim

The U.S. facilitated the sending of nearly 17.8 tons of donated medical supplies to China to combat the spread of the novel coronavirus in early 2020.

Rating



True

[About this rating](#)

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AD - ARTICLE CONTINUES BELOW

Origin

*As governments fight the COVID-19 pandemic, Snopes is **fighting** an “infodemic” of rumors and misinformation, and you can help. **Read** our coronavirus fact checks. **Submit** any questionable rumors and “advice” you encounter. **Become a Founding Member** to help us hire more fact-checkers. And, please, follow the **CDC** or **WHO** for guidance on protecting your community from the disease.*

During the COVID-19 coronavirus disease pandemic in March 2020, social media users began **sharing** a tweet ostensibly posted by U.S. Rep. Maxine Waters (D-CA) that



Did Bill Gates Admit Vaccinations Are Designed So Governments Can Depopulate the World?



Does Ivanka Trump Hold a Trademark on Coffins in China?



Did Michigan's Governor Ban the Sale of American Flags?



Is Bill Gates Being Sued by India Over Vaccination Deaths?

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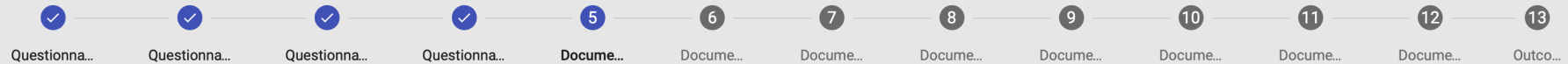
[View all](#)

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Claimant/Source: Facebook Post

Date: 2020-02-04

Statement: On February 7, the WHO warned about the limited stock of PPE. That same day, the Trump administration announced it was sending 18 tons of masks, gowns and respirators to China.

Please select one of the URLs below to justify the truthfulness value of the statement, then input the truth level as one from the six labels provided and write your justification.

Insert your query

February 7 WHO limited stock of PPE trump

SEARCH

46 results found.

[Did the Trump Administration Send 18 Tons of PPE to China ...](https://www.snopes.com/fact-check/us-tons-ppe-china/)

<https://www.snopes.com/fact-check/us-tons-ppe-china/>

Did the Trump Administration Send 18 Tons of PPE to China in Early 2020? ... of PPE are not new. Back on February 7, the World Health Organization sounded alarm bells about "the limited stock of ...

Select ☒

[Trump administration sent protective medical gear to China ...](https://www.msn.com/en-us/health/medical/trump-administration-sent-protective-medical-gear-to-china-while-he-was-minimizing-the-virus-threat-to-us/ar-BB11SSuo)

<https://www.msn.com/en-us/health/medical/trump-administration-sent-protective-medical-gear-to-china-while-he-was-minimizing-the-virus-threat-to-us/ar-BB11SSuo>

Concerns about a dwindling supply of PPE are not new. Back on February 7, the World Health Organization sounded alarm bells about "the limited stock of PPE," noting demand was 100 times higher ...

Select ☐

[Trump admin sent medical gear to China while he minimized ...](https://dnyuz.com/2020/03/30/trump-admin-sent-medical-gear-to-china-while-he-minimized-the-virus-threat-to-us/)

<https://dnyuz.com/2020/03/30/trump-admin-sent-medical-gear-to-china-while-he-minimized-the-virus-threat-to-us/>

Concerns about a dwindling supply of PPE are not new. Back on February 7, the World Health Organization sounded alarm bells about "the limited stock of PPE," noting demand was 100 times higher than normal for this equipment.. Yet the same day as the WHO warning, the Trump administration announced that it was transporting to China nearly 17.8 tons (more than 35,000 pounds) of "masks ...

Select ☐

[America Has PPE Shortage Because Trump Sent Ours to China ...](https://www.themarysue.com/shortage-of-ppe-because-trump-sent-our-stockpile-to-china/)

<https://www.themarysue.com/shortage-of-ppe-because-trump-sent-our-stockpile-to-china/>

America Has a Shortage of PPE Because Trump Sent Our Stockpile to China Among all of Trump's poor decisions, this might be the deadliest.

Select ☐

[Reports that Trump sent PPE equipment to China Feb 4 ...](https://2peasrefugees.boards.net/thread/104967/reports-trump-sent-equipment-china)

<https://2peasrefugees.boards.net/thread/104967/reports-trump-sent-equipment-china>

There are reports that Trump sent 17.8 tons of PPE equipment to China on February 4. (tweet from Pompeo @secretary Pompeo February 7. There are also reports that Ottawa

Select ☐

[Reports that Trump sent PPE equipment to China Feb 4 ...](#)

<https://2peasrefugees.boards.net/thread/104967/reports-trump-sent-equipment-china>

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Select ☐

[Sick of the Slant - A Fair Look at the News of the Day ...](#)

<https://www.facebook.com/pg/SickOfTheSlant/posts/>

(Albert) "On February 7, the (World Health Organization) warned about the limited stock of PPE. That same day, the Trump administration announced it was sending 18 tons of masks, gowns and respirators to China.

Select ☐

[CNN - Breaking News, Latest News and Videos](#)

https://lite.cnn.com/en/article/h_c9ec764aa333434b8543e28250957b40

Back on February 7, the World Health Organization sounded alarm bells about "the limited stock of PPE," noting demand was 100 times higher than normal for this equipment. Yet the same day as the WHO warning, the Trump administration announced that it was transporting to China nearly 17.8 tons (more than 35,000 pounds) of "masks, gowns, gauze ...

Select ☐

[Why did Trump send 18 tons of medical equipment to China...](#)

<https://247sports.com/college/texas-tech/board/102960/Contents/why-did-trump-send-18-tons-of-medical-equipment-to-china-145529702/?page=1>

Why did Trump send 18 tons of medical equipment to China...

Select ☐

[Trump Sends 35,000 Pounds of PPE to China | GopherHole ...](#)

<https://www.forums.gopherhole.com/boards/threads/trump-sends-35-000-pounds-of-ppe-to-china.95940/>

Trump Sends 35,000 Pounds of PPE to China. Thread starter stocker08; Start date 25 minutes ago; stocker08 Well-known member. Joined Feb 2, 2009 Messages 18,001 Reaction score 261 Points 83. 25 minutes ago #1 Why is this being glossed over? Trump has been a complete fool throughout this entire debacle. He's minimized this every step of the way ...

Select ☐

[US Shipping China Covid 19 Medical Supplies?](#)

<https://www.debatepolitics.com/covid-19-coronavirus-disease/392029-us-shipping-china-covid-19-medical-supplies.html>

Concerns about a dwindling supply of PPE are not new. Back on February 7, the World Health Organization sounded alarm bells about "the limited stock of PPE," noting demand was 100 times higher than normal for this equipment. Yet the same day as the WHO warning, the Trump administration announced that it was transporting to China nearly 17.8 ...

Select ☐

Items per page: 10

1 - 10 of 46



- ☐ Lie
- ☐ False
- ☐ Barely True
- ☐ Half True
- ☐ Mostly True
- ☒ True

Your justification here

NEXT

(*) you have to fill each field to proceed

[Reports that Trump sent PPE equipment to China Feb 4 ...](#)

<https://2peasrefugees.boards.net/thread/104967/reports-trump-sent-equipment-china>

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<https://www.facebook.com/pg/SickOfTheSlant/posts/>

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[CNN - Breaking News, Latest News and Videos](#)

https://lite.cnn.com/en/article/h_c9ec764aa333434b8543e28250957b40

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Select ☐

[Why did Trump send 18 tons of medical equipment to China...](#)

<https://247sports.com/college/texas-tech/board/102960/Contents/why-did-trump-send-18-tons-of-medical-equipment-to-china-145529702/?page=1>

Why did Trump send 18 tons of medical equipment to China...

Select ☐

[Trump Sends 35,000 Pounds of PPE to China | GopherHole ...](#)

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Concerns about a dwindling supply of PPE are not new. Back on February 7, the World Health Organization sounded alarm bells about "the limited stock of PPE," noting demand was 100 times higher than normal for this equipment. Yet the same day as the WHO warning, the Trump administration announced that it was transporting to China nearly 17.8 ...

Select ☐

Items per page: 10

1 - 10 of 46



- ☐ Lie
- ☐ False
- ☐ Barely True
- ☐ Half True
- ☐ Mostly True
- ☒ True

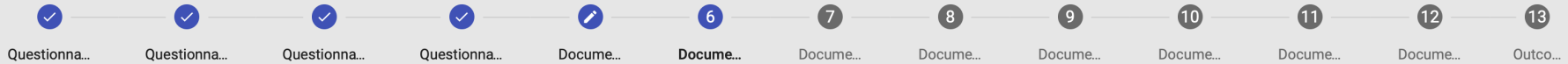
[Your justification here](#)

The U.S. facilitated the sending of nearly 17.8 tons of donated medical supplies to China to combat the spread of the novel coronavirus in early 2020.



NEXT

(*) you have to fill each field to proceed



Please judge the truthfulness of the following statement

Claimant/Source: Abraham Lincoln

Date: 2020-04-01

Statement: If I were not a Formula One pilot, I would be a football player.

Please select one of the URLs below to justify the truthfulness value of the statement, then input the truth level as one from the six labels provided and write your justification.

Insert your query

Abraham Lincoln

SEARCH

49 results found.

[Abraham Lincoln - Wikipedia](https://en.wikipedia.org/wiki/Abraham_Lincoln)

https://en.wikipedia.org/wiki/Abraham_Lincoln

Abraham Lincoln was born on February 12, 1809, the second child of Thomas Lincoln and Nancy Hanks Lincoln, in a one-room log cabin on Sinking Spring Farm near Hodgenville, Kentucky.: 20–22 He was a descendant of Samuel Lincoln, an Englishman who migrated from Hingham, Norfolk, to its namesake, Hingham, Massachusetts, in 1638.

Select ☒

[Abraham Lincoln - Facts, Birthday & Assassination - HISTORY](https://www.history.com/topics/us-presidents/abraham-lincoln)

<https://www.history.com/topics/us-presidents/abraham-lincoln>

Abraham Lincoln, a self-taught lawyer, legislator and vocal opponent of slavery, was elected 16th president of the United States in November 1860, shortly before the outbreak of the Civil War. He ...

Select ☐

[Abraham Lincoln | Biography, Facts, History, & Childhood ...](https://www.britannica.com/biography/Abraham-Lincoln)

<https://www.britannica.com/biography/Abraham-Lincoln>

Abraham Lincoln was a member of the Whig Party and later a Republican. He believed that the government's job was to do what a community of people could not do for themselves. One of his greatest preoccupations as a political thinker was the issue of self-governance and the promise and problems that could arise from it.

Select ☐

[Abraham Lincoln | The White House](https://www.whitehouse.gov/about-the-white-house/presidents/abraham-lincoln/)

<https://www.whitehouse.gov/about-the-white-house/presidents/abraham-lincoln/>

Abraham Lincoln became the United States' 16th President in 1861, issuing the Emancipation Proclamation that declared forever free those slaves within the Confederacy in 1863. Lincoln warned the ...

Select ☐

[Abraham Lincoln Facts for Kids](https://kids.kiddle.co/Abraham_Lincoln)

https://kids.kiddle.co/Abraham_Lincoln

Abraham Lincoln (February 12 1809 – April 15 1865) was the 16th President of the United States. He served as president from 1861 to 1865, during the American Civil War. Just five days after most of the Confederate forces had surrendered and the war was ending, John Wilkes Booth assassinated Lincoln.. Lincoln was the first president of the United States to be assassinated.

Select ☐

[Abraham Lincoln Facts for Kids](#)

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Abraham Lincoln (February 12 1809 – April 15 1865) was the 16th President of the United States.He served as president from 1861 to 1865, during the American Civil War.Just five days after most of the Confederate forces had surrendered and the war was ending, John Wilkes Booth assassinated Lincoln.. Lincoln was the first president of the United States to be assassinated.

Select ☐

[TOP 25 QUOTES BY ABRAHAM LINCOLN \(of 1141\) | A-Z Quotes](#)

https://www.azquotes.com/author/8880-Abraham_Lincoln

"Abraham Lincoln: Speeches & Writings Part 2: 1859-1865: Library of America #46", p.627, Library of America 1 Copy quote. I do the very best I can, I mean to keep going. If the end brings me out all right, then what is said against me won't matter. If I'm wrong, ten angels swearing I was right won't make a difference.

Select ☐

[Biography of Abraham Lincoln, 16th U.S. President](#)

<https://www.thoughtco.com/abraham-lincoln-16th-president-united-states-104273>

Abraham Lincoln (February 12, 1809–April 15, 1865) was the 16th president of the United States, serving from 1861 to 1865. During his time in office, the nation fought the Civil War, which claimed hundreds of thousands of lives. One of Lincoln's greatest accomplishments was the abolition of slavery in 1864.

Select ☐

[50 interesting facts about Abraham Lincoln's life ...](#)

<https://constitutioncenter.org/blog/50-shades-of-abraham-lincoln-2/>

Abraham Lincoln, the 16th president, has a birthday today. If you are a Lincoln fan, here are some cool facts, including Lincoln's career as an inventor, his love of animals, and his one losing appearance before the Supreme Court.

Select ☐

[Abraham Lincoln's Assassination - HISTORY](#)

<https://www.history.com/topics/american-civil-war/abraham-lincoln-assassination>

Abraham Lincoln, a self-taught lawyer, legislator and vocal opponent of slavery, was elected 16th president of the United States in November 1860, shortly before the outbreak of ...

Select ☐

[Abraham Lincoln - Simple English Wikipedia, the free ...](#)

https://simple.wikipedia.org/wiki/Abraham_Lincoln

Abraham Lincoln (February 12, 1809 – April 15, 1865) was an American politician. He was the 16th President of the United States.He was president from 1861 to 1865, during the American Civil War.Just five days after most of the Confederate forces had surrendered and the war was ending, John Wilkes Booth assassinated Lincoln for being a tyrant.Lincoln was the first president of the United ...

Select ☐

Items per page: 10

1 – 10 of 49



- ☒ Lie
- ☐ False
- ☐ Barely True
- ☐ Half True
- ☐ Mostly True
- ☐ True

[Your justification here](#)

Abraham Lincoln was born on February 12, 1809,



BACK

NEXT

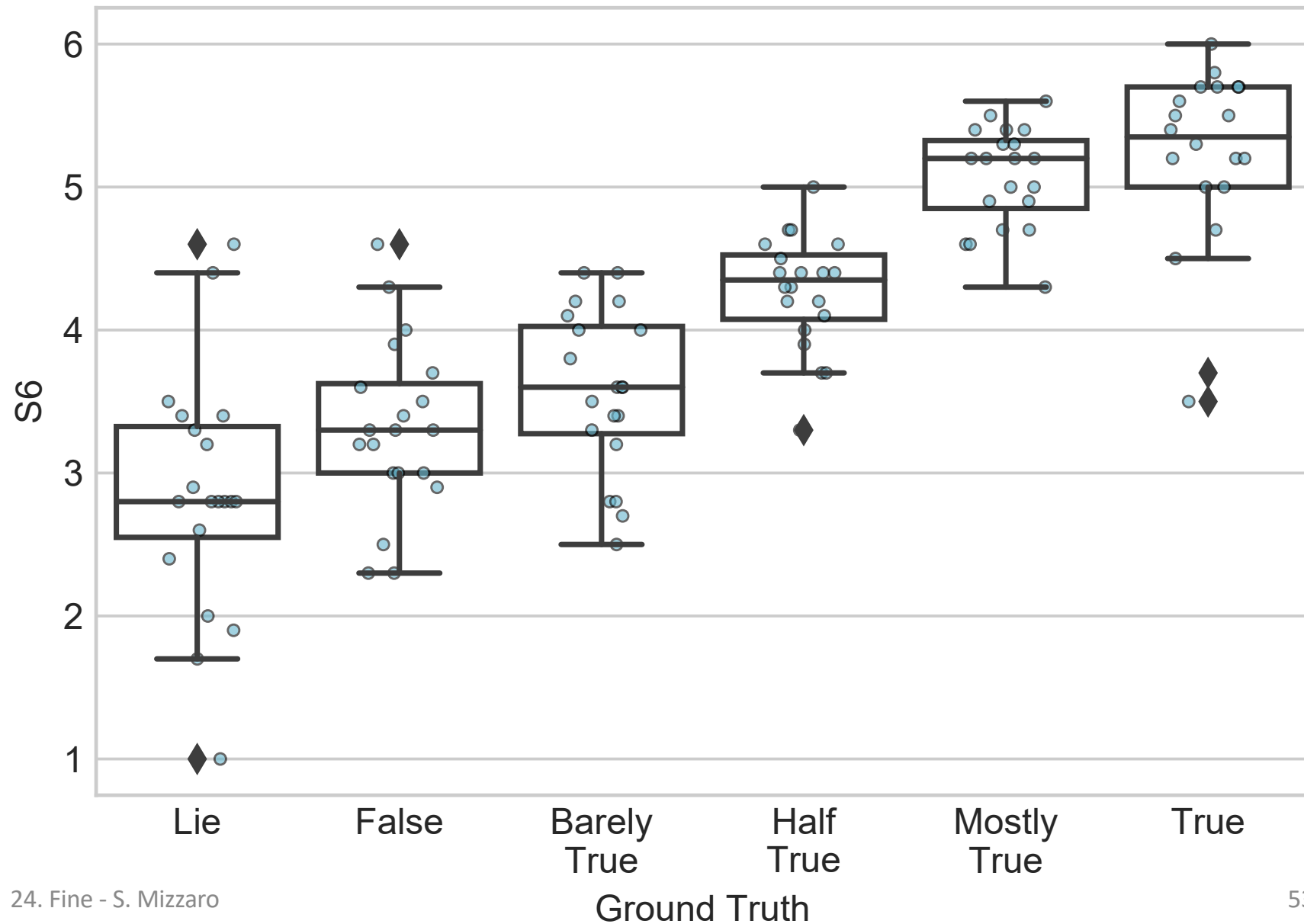
(*) you have to fill each field to proceed

Controlli di qualità

- CRT (le tre domande iniziali)
- Due "Gold question" (una True e una Lie)
- Monitoraggio
 - Tempi
 - Azioni
- URL e testi inseriti
- ...
- Ragionevolmente "sicuri" che i worker facevano un lavoro di qualità

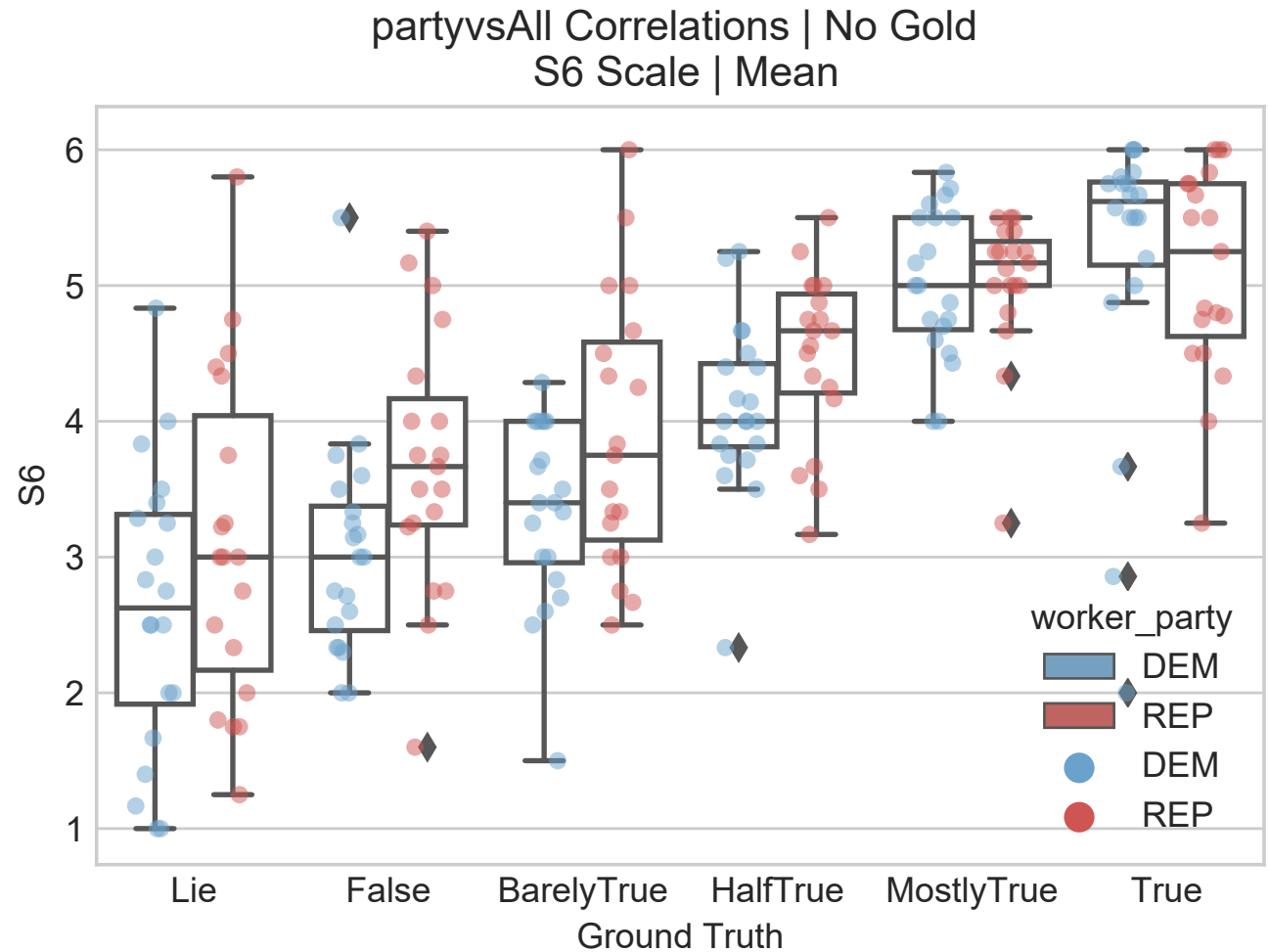
Risultati (alcuni)

Accordo con "Ground truth" [ECIR 20]



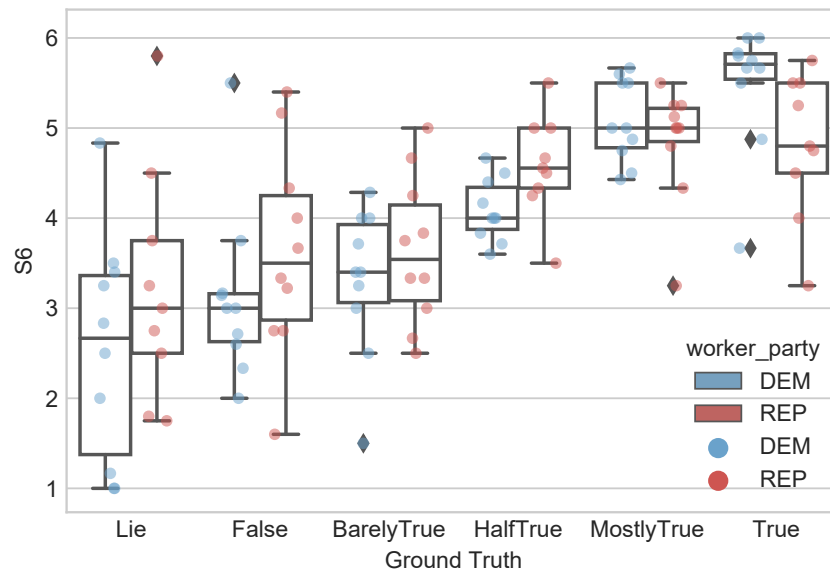
Background [ECIR 20]

- Repubblicani danno voti più alti
 - I pallini rossi più in alto dei blu

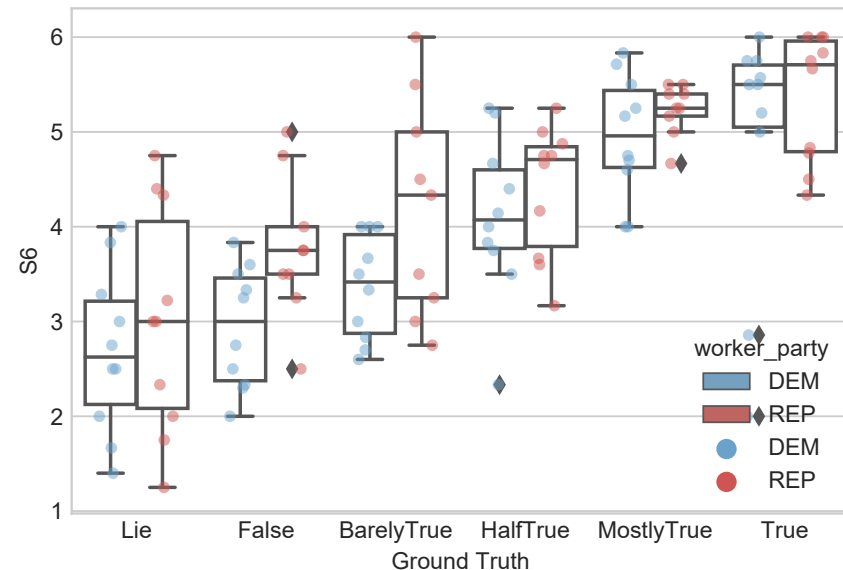


Bias, "pregiudizio" [ECIR 20]

Affermazioni di Democratici

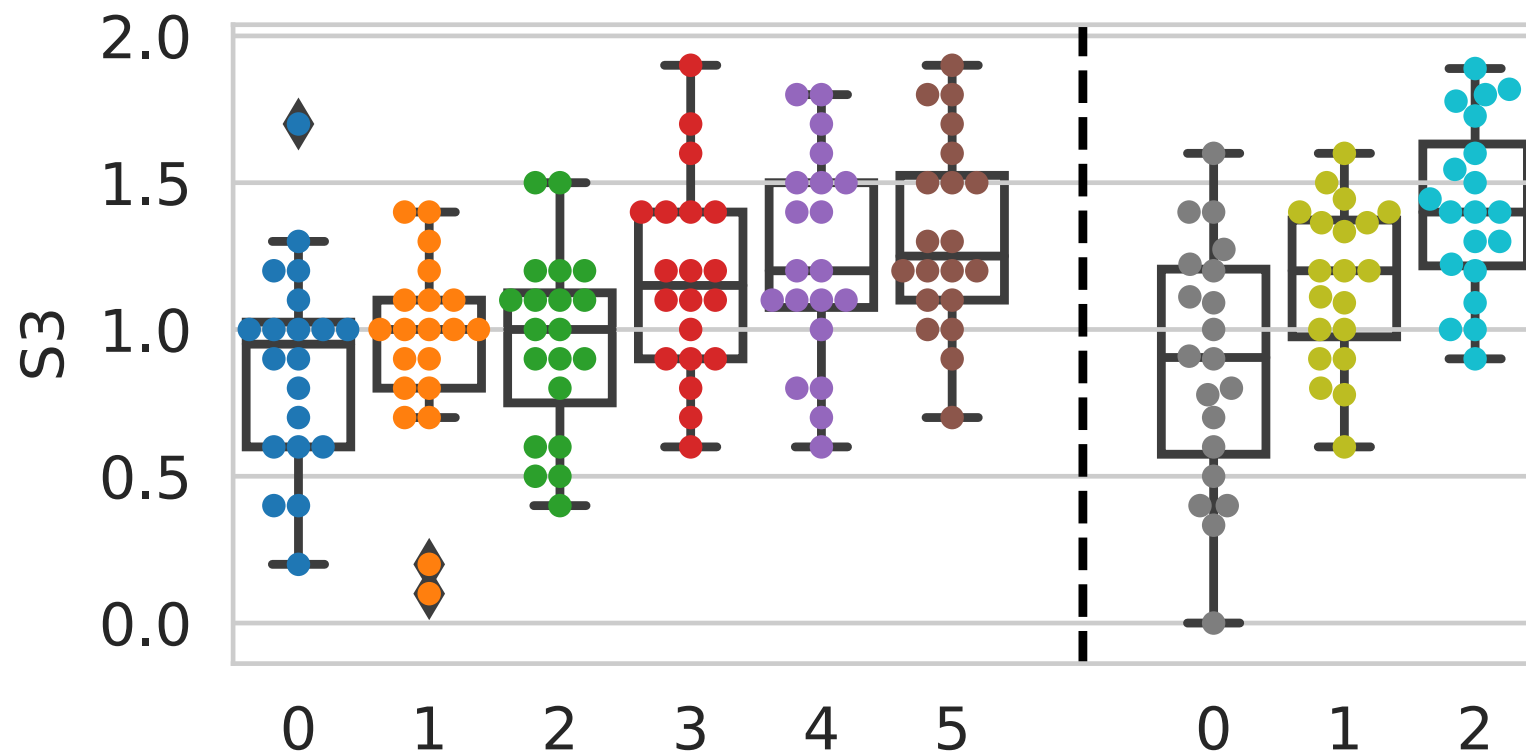


Affermazioni di Repubblicani

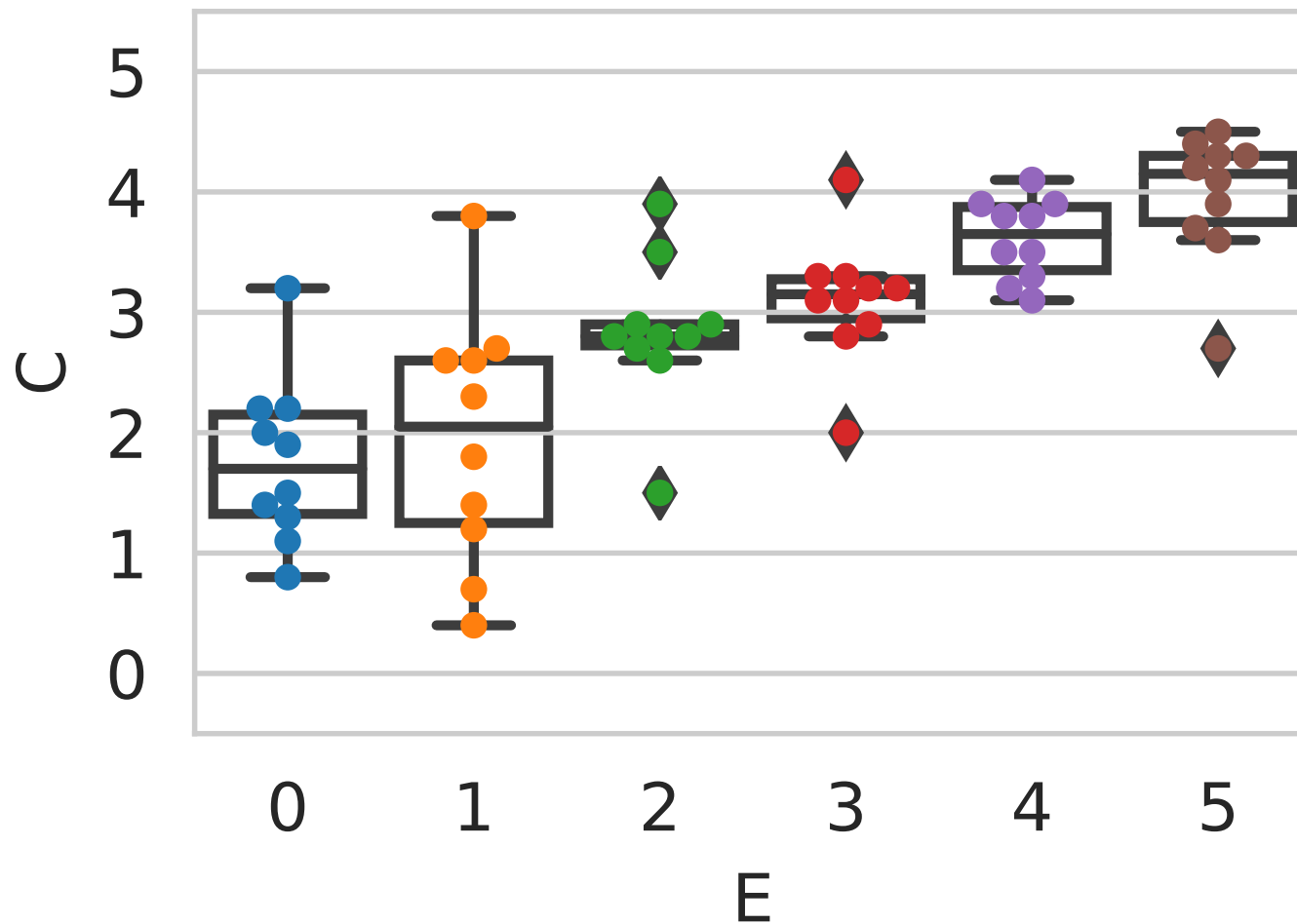


- Bias "partigiano", più forte per i Repubblicani

Un altro risultato [SIGIR 20, fig. 2]



Un altro [CIKM 20, fig. 2]



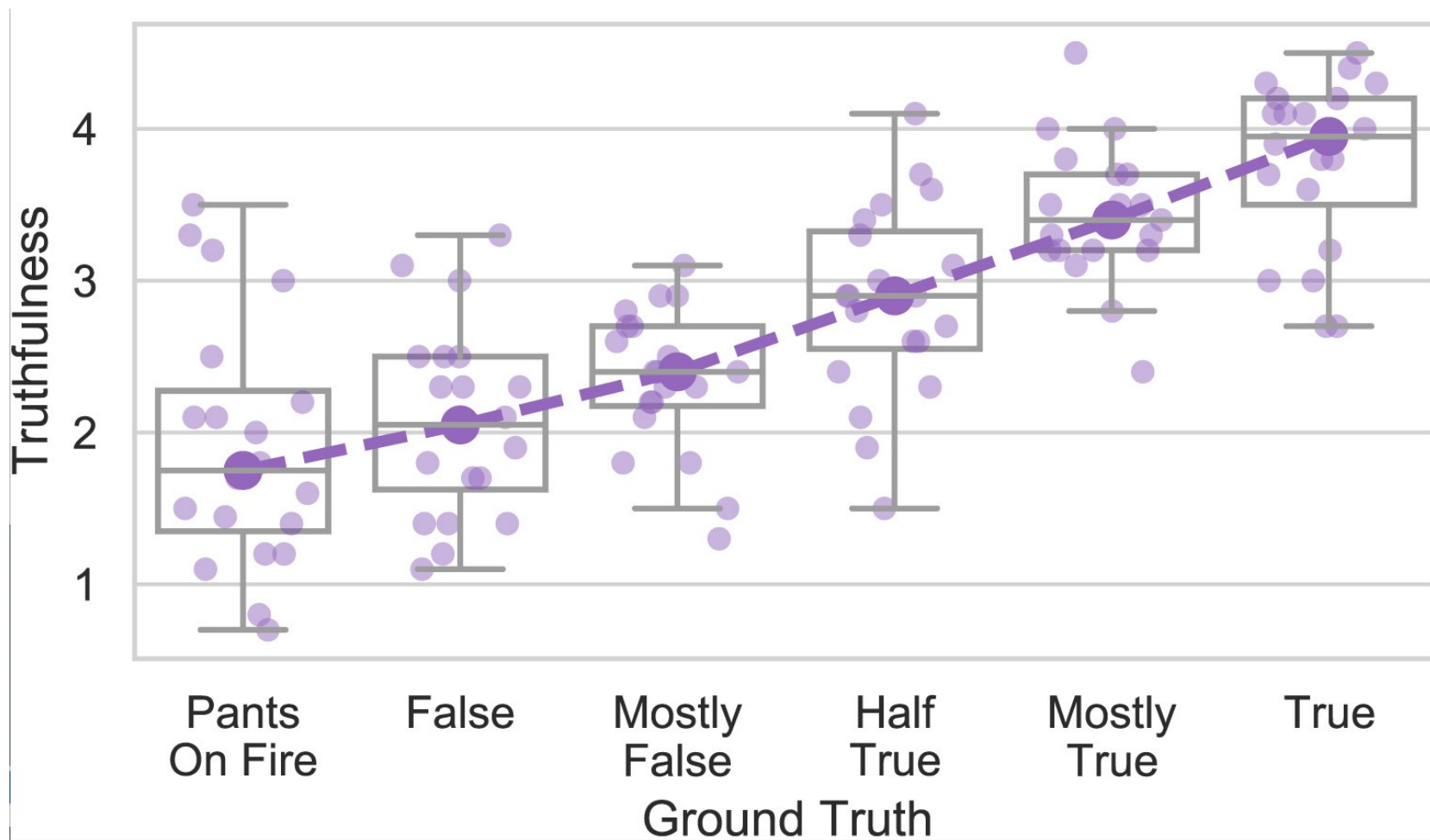
Insomma

- C'è un segnale 😊
- Ma ci sono anche errori 😞
- "Binarizzando" le categorie (True/False) si ha un'accuracy di ca. il 70% 🤔
- Analoga a quella dei sistemi automatici 😬
 - (il che è un problema!)
 - Tutta 'sta fatica per niente?!

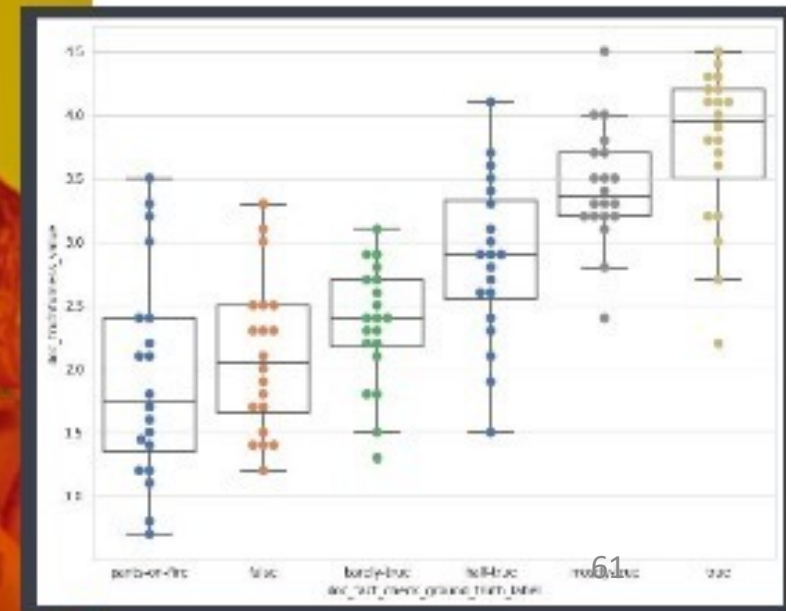
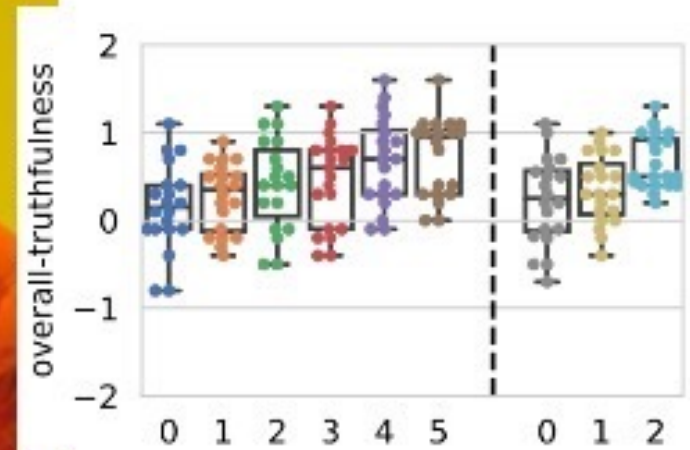
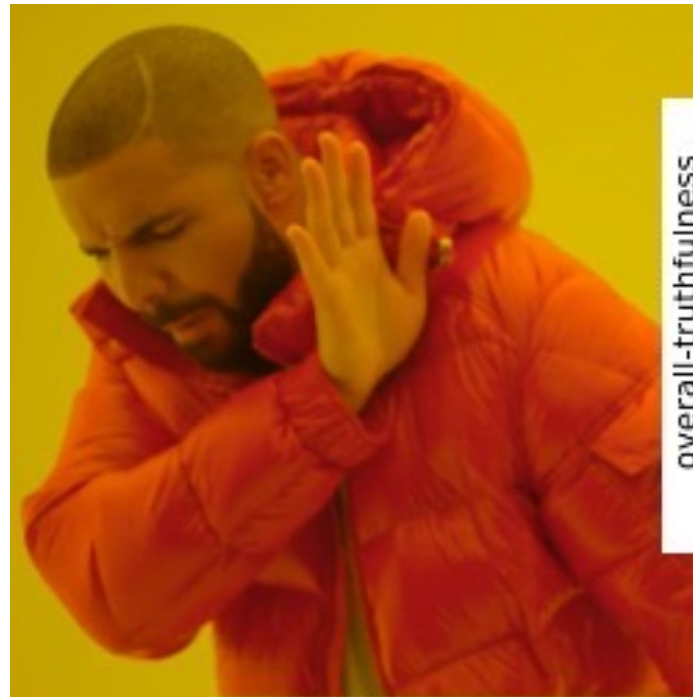
Ci abbiamo riprovato

- [IP&M 2024b]
- Esperimento simile
- Piccole migliorie varie
 - (istruzioni, testi, formati, etichette, ...)
- Su Prolific
 - Altra piattaforma, qualità > di mturk

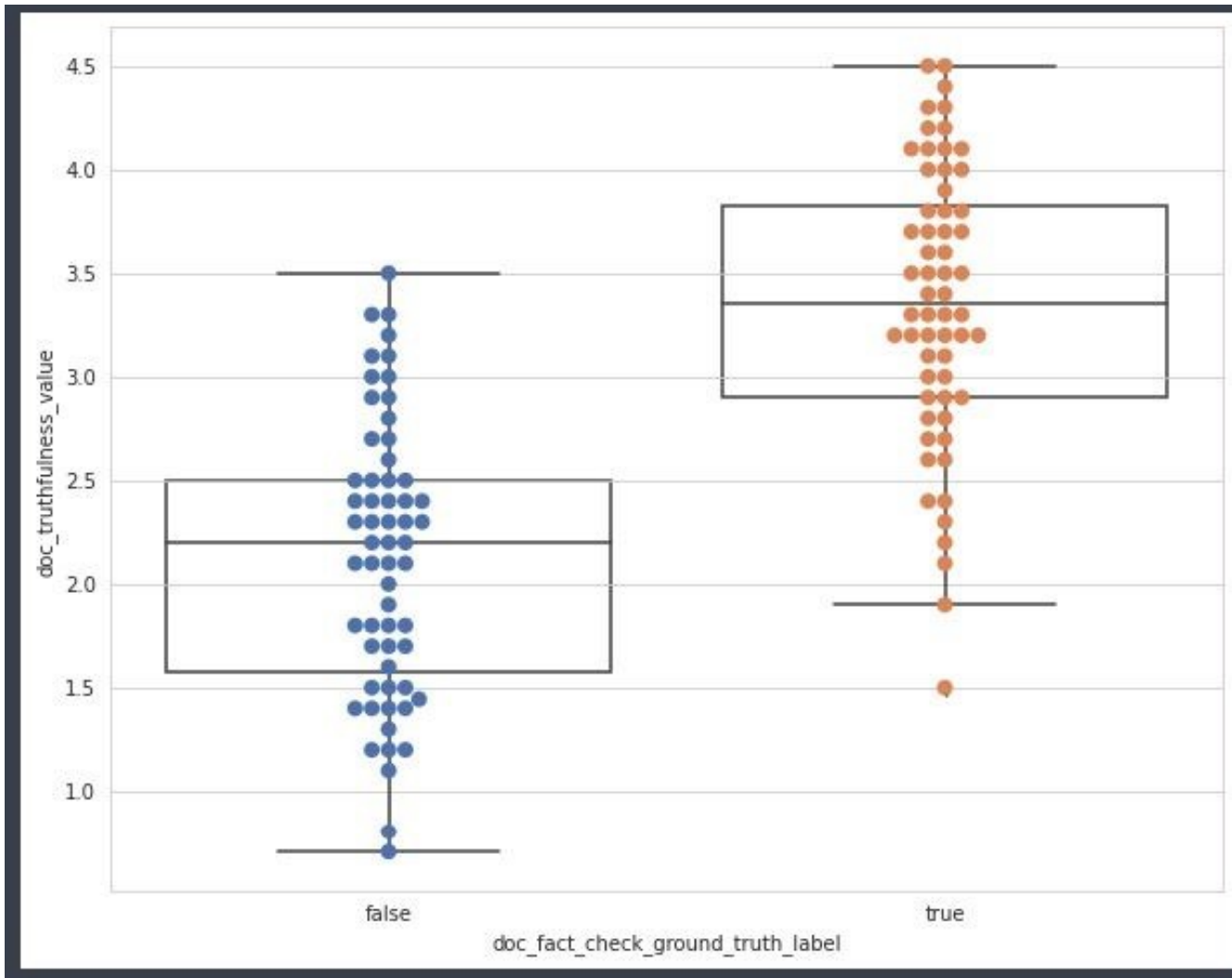
Risultato principale



1a reazione

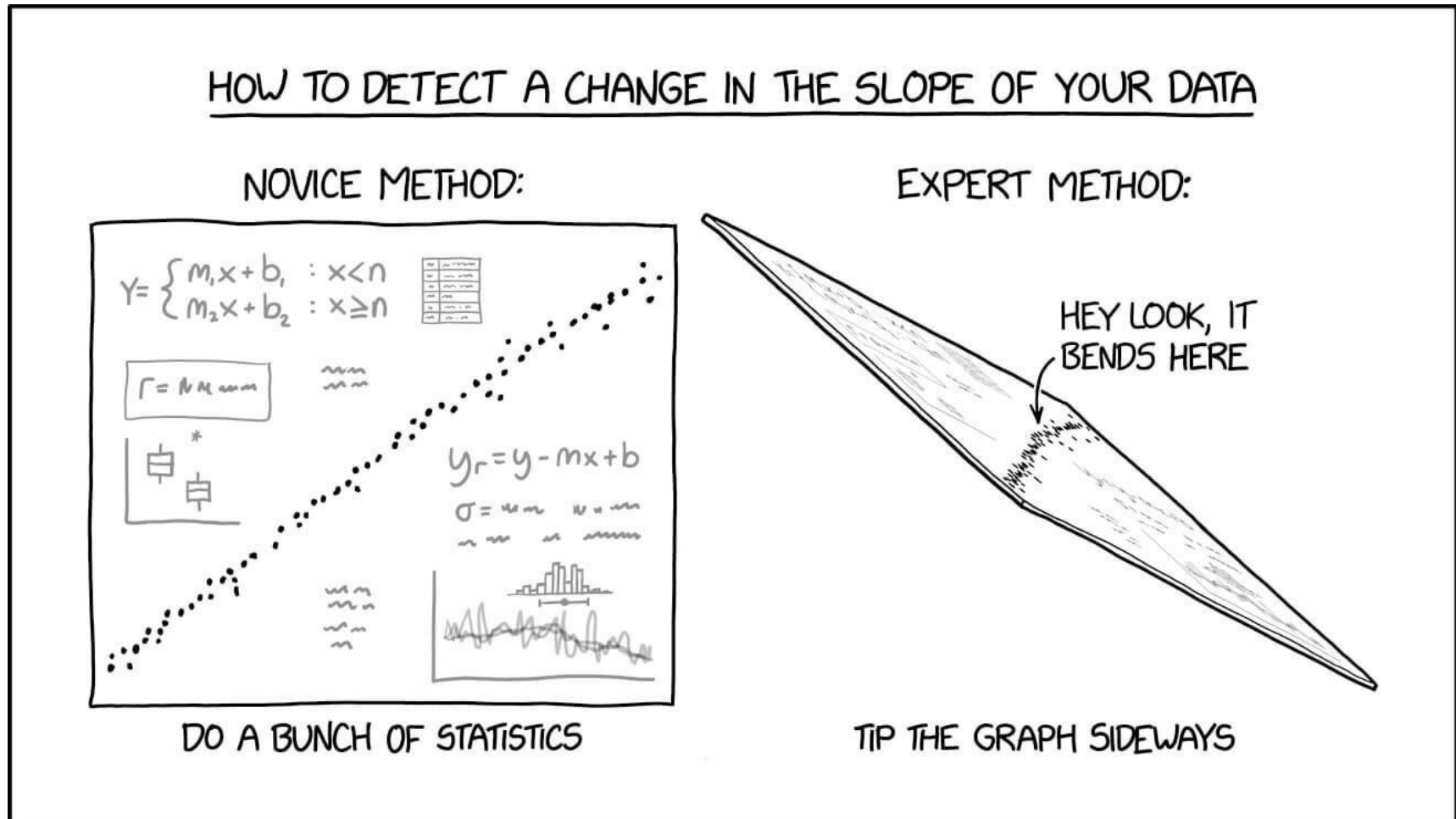


Però poi...

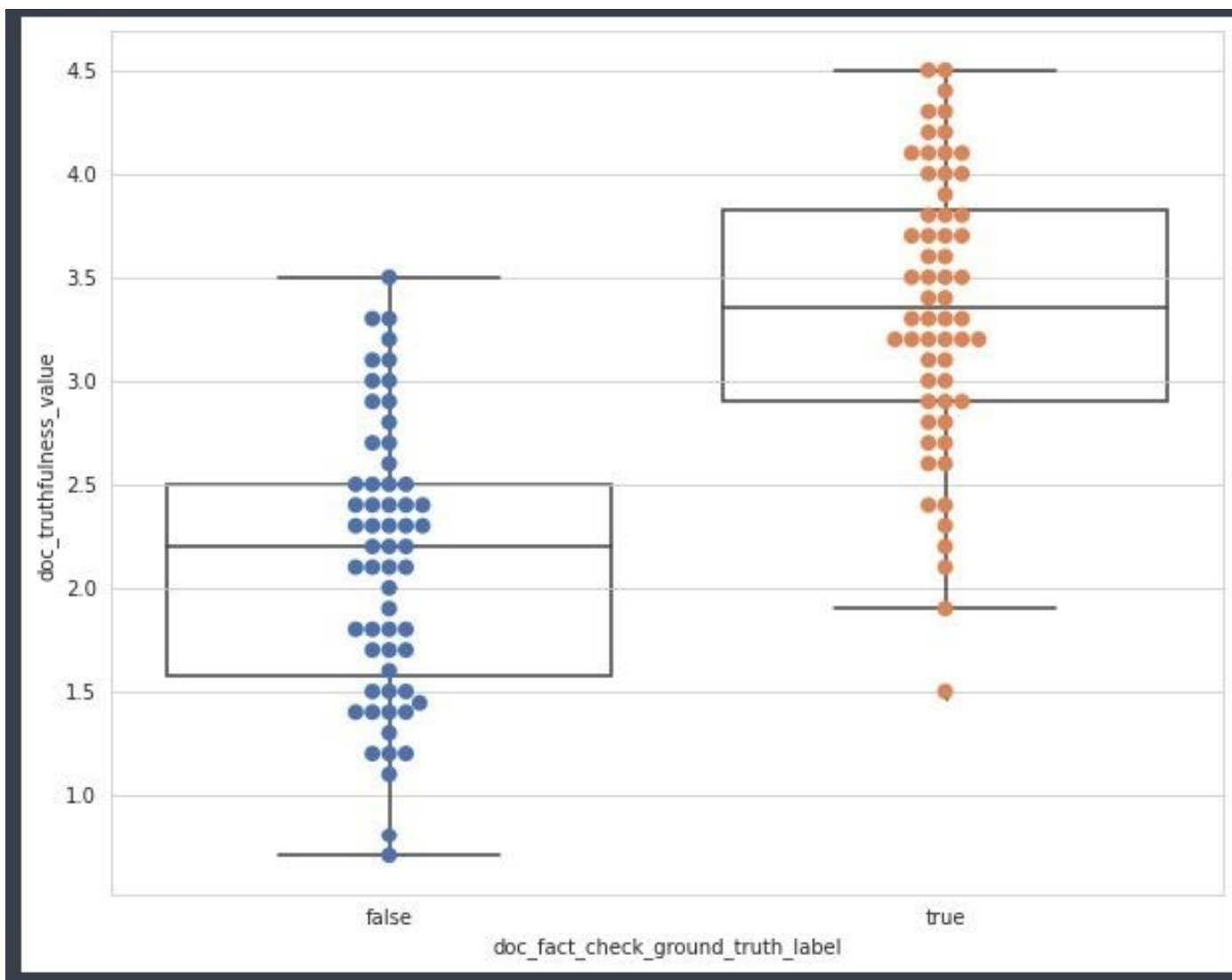


- Però...
- E binarizzando quant'è l'accuracy?
- «Accuracy = 0.7» 😞
- Agh
- Non meglio del ML

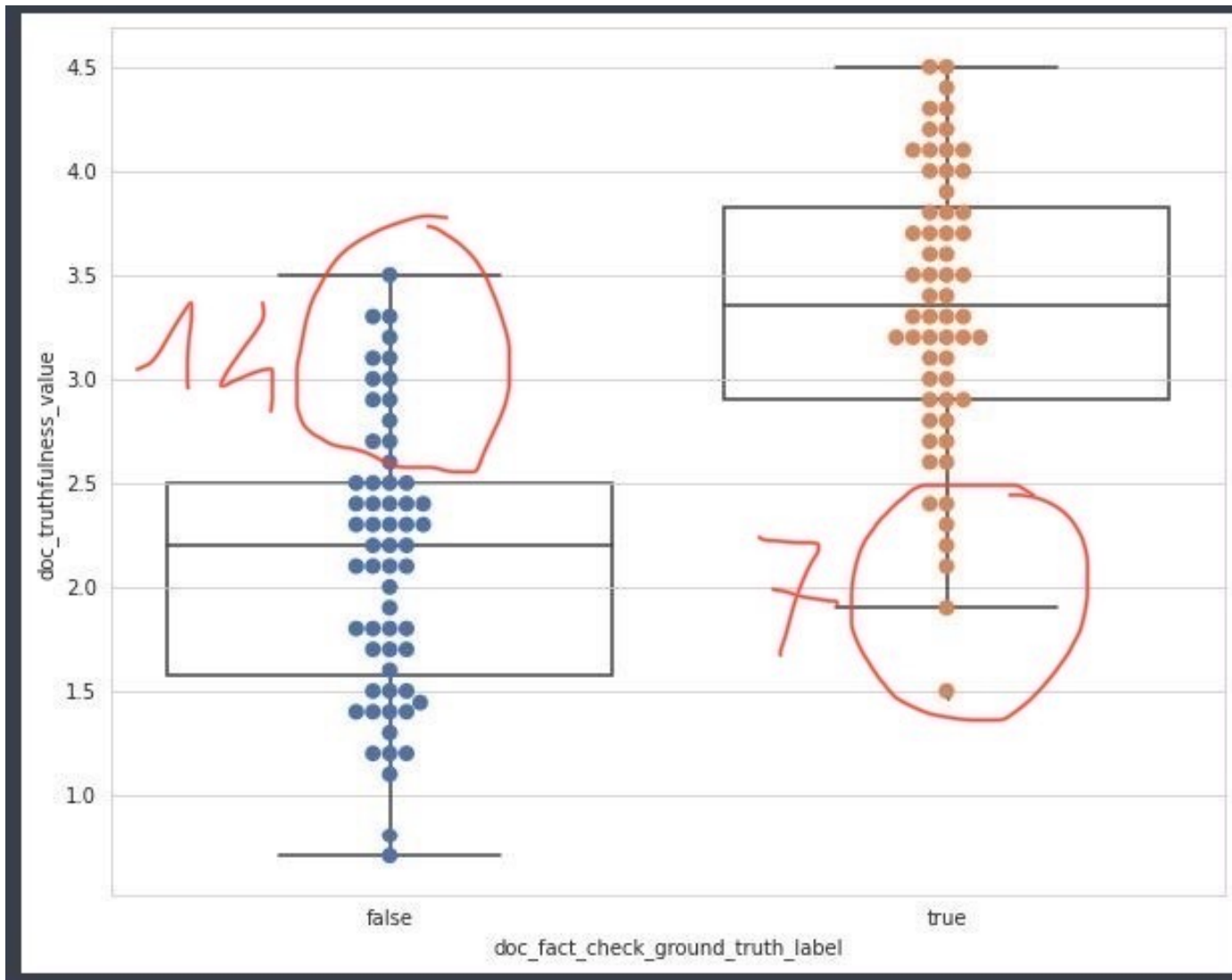
...ma:



Ho contato i puntini



Ho contato i puntini



Accuracy

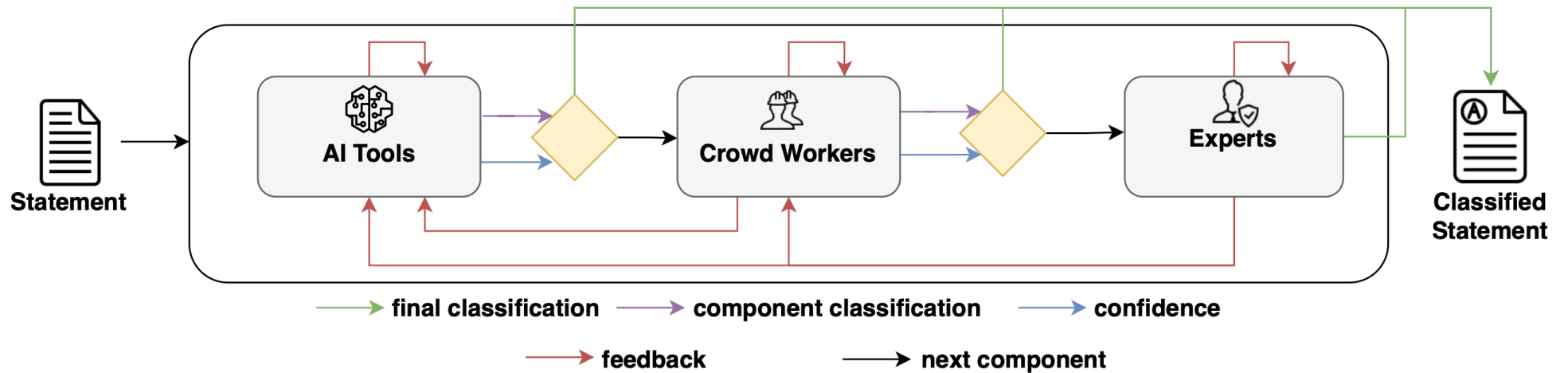
- $60 - 14 = 46$
- $60 - 7 = 53$
- $46 + 53 = 99$
- $99 / 120 = 0.82$
- 0.7? Eh no, 0.8!
- ...discussioni... bug nel codice
 - (" ≤ 2 e ≥ 3 "...)
- > di quella del ML, finalmente

Quindi

- Anche se compito difficile per il crowd
- Risultati sensati
- Ma non ancora ottimali
 - Errori sugli statement "in alto a sinistra e in basso a destra"
- Ci stiamo lavorando
 - Scala multidimensionale
 - Worker più di qualità
 - Aggregazione più sofisticata
 - Failure analysis
 - Sistemi ibridi Crowd + AI
 - ...
- Possibili tesi

Obiettivo: Sistema ibrido HITL

- HITL = Human-In-The-Loop



Il team

- Isabelle Augenstein
- Francesco Bombassei De Bona
- Davide Ceolin
- Alessandro Checco
- Vincenzo Della Mea
- Gianluca Demartini
- Massimiliano De Luise
- Tim Draws
- Shaoyang Fan
- David La Barbera
- Joel Mackenzie
- Eddy Maddalena
- Beatrice Portelli
- Yunke Qu
- Kevin Roitero
- Giuseppe Serra
- Michael Soprano
- Damiano Spina
- Denis Eduard Tapu
- Dustin Wright
- Arkaitz Zubiaga

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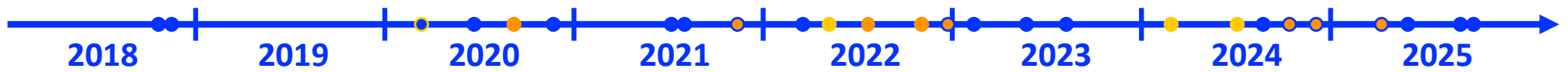
Tante persone, tanti lavori

- 7 – 8 anni, 22 persone → 26 articoli
 - (almeno)
- Progetti di ricerca
 - Facebook
 - PRIN
- ...

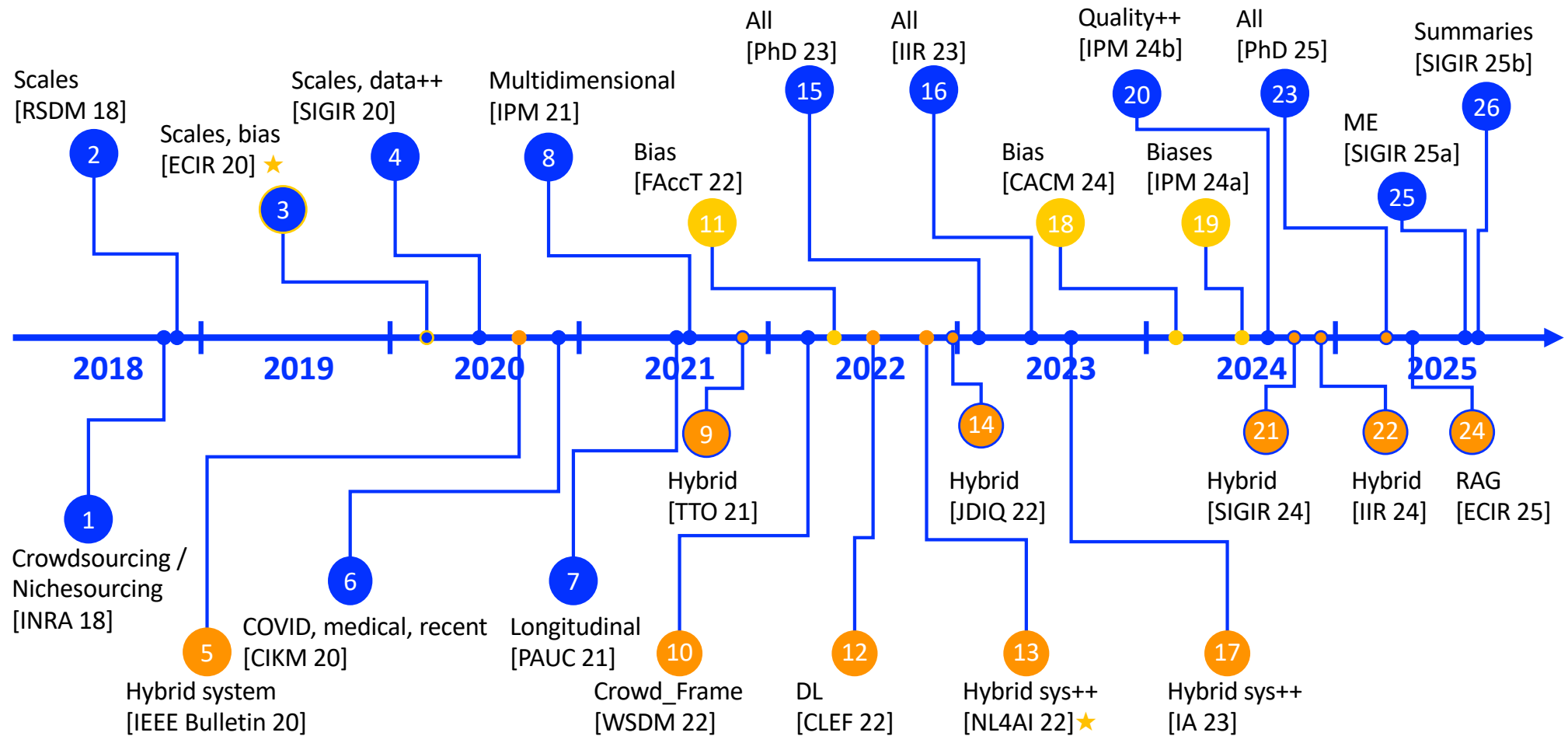
Timeline



Timeline



Timeline



Biblio – Our papers, grouped

- Several similar experiments
 - 1. Crowdsourcing vs. Nichesourcing [INRA 2018]
 - 2. Scales [RDSM 2018]
 - 3. Initial, a few data, worker background, scales, bias [ECIR 2020] ★
 - 4. More data, scales [SIGIR 2020]
 - 6. COVID-19: medical, recent news [CIKM 2020]
 - 7. Longitudinal: [PAUC 2021]
 - 8. Multidimensional: more aspects [IPM 2021]
 - 20. Quality > (Prolific, UI, ...) [IPM 2024b]
 - 25. ME [SIGIR 2025a]
 - 26. Summaries [SIGIR 2025b]
 - Implementation – Hybrid system
 - 5. First proposal [Bulletin 2020]
 - 10. Crowd_Frame [WSDM 2022]
 - 12. Deep Learning [CLEF 2022]
 - 13. System description [NL4AI2022] ★
 - 17. Hybrid system ++ [IA 2023]
 - Hybrid approaches Crowd + LLMs
 - 9. First experiments [TTO 2021]
 - 14. Hybrid [JDIQ 2022]
 - 21. Hybrid++ [SIGIR 2024]
 - 22. Hybrid [IIR 2024]
 - 24. RAG [ECIR 2025]
 - Bias
 - 11. Bias [FAccT 2022]
 - 18. Bias management [CACM 2023]
 - 19. Biases in Fact-checking [IPM 2024a]
 - All
 - 15. PhD Thesis Soprano [PhD 2023]
 - 16. Summary [IIR 2023]
 - 23. PhD Thesis La Barbera [PhD 2025]
- (details in next slides)

1. [INRA 2018]

- E. Maddalena, D. Ceolin, S. Mizzaro.
"Multidimensional News Quality: A Comparison of Crowdsourcing and Nichesourcing." In *CIKM Workshops -- Proceedings of the 6th International Workshop on News Recommendation and Analytics (INRA 2018)*, 2018.
- <http://ceur-ws.org/Vol-2482/paper17.pdf>

2. [RDSM 2018]

- K. Roitero, G. Demartini, S. Mizzaro, D. Spina. How Many Truth Levels? Six? One Hundred? Even More? Validating Truthfulness of Statements via Crowdsourcing. In *CIKM Workshops -- 2nd International Workshop on Rumours and Deception in Social Media (RDSM)*. 2018.
- <https://www.damianospina.com/publication/roitero-2018-how/roitero-2018-how.pdf>

3. [ECIR 2020]

- D. La Barbera, K. Roitero, G. Demartini, S. Mizzaro, D. Spina. Crowdsourcing Truthfulness: The Impact of Judgment Scale and Assessor Bias. Advances in Information Retrieval - 42nd European Conference on IR Research, ECIR 2020. 207-214
- Video:

<https://www.youtube.com/watch?v=9wFFMcplvjk>



ECIR 2020 @ecir2020 · 1g

#ECIR2020 Best **Short Paper Award** goes to "Crowdsourcing Truthfulness; The Impact of Judgment Scale and Assessor Bias" authored by

David la Barbera, Kevin Roitero, Damiano Spina, Stefano Mizzaro, Gianluca Demartini. Congratulations!!



4. [SIGIR 2020]

- K. Roitero, M. Soprano, S. Fan, D. Spina, S. Mizzaro, G. Demartini. Can The Crowd Identify Misinformation Objectively? The Effects of Judgment Scale and Assessor's Background. Proceedings of the 43st International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR 2020). Xi'an, China (Online). July 25-30, 2020, <https://doi.org/10.1145/3397271.3401112>
- Video: <https://www.youtube.com/watch?v=D10EtrThvbc>

5. [Bulletin 2020]

- G. Demartini, S. Mizzaro, D. Spina. Human-in-the-loop artificial intelligence for fighting online misinformation: challenges and opportunities. *Bulletin of the IEEE Computer Society Technical Committee on Data Engineering* 43.3 (2020): 65-74.
- <https://www.damianospina.com/publication/demartini-2020-human/demartini-2020-human.pdf>

6. [CIKM 2020]

- K. Roitero, M. Soprano, B. Portelli, D. Spina, V. Della Mea, G. Serra, S. Mizzaro, G. Demartini. The COVID-19 Infodemic: Can the Crowd Judge Recent Misinformation Objectively?, Proceedings of the 29th ACM International Conference on Information and Knowledge Management (CIKM 2020). Galway, Ireland (Online). October 19-23, 2020, <https://doi.org/10.1145/3340531.3412048>
- Video: <https://dl.acm.org/doi/10.1145/3340531.3412048>

7. [PAUC 2021]

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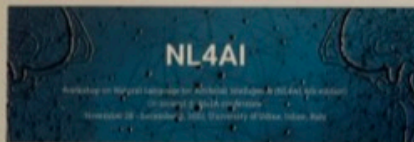
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Associazione
Italiana per
l'Intelligenza
Artificiale

CERTIFICATE

FOR BEST PAPER

NL4AI 2022

This certificate is proudly awarded by

David La Barbera, Kevin Roitero and Stefano Mizzaro
"A Hybrid Human-In-The-Loop Framework for Fact Checking"



A handwritten signature in black ink, likely belonging to one of the chairs of the competition.

Chairs

Debora Nozza, Lucia Passaro, Marco Polignano

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Scaletta

- Fine accordo
- Fake news
- **Riassunto corso**
- Esame
 - Regole
 - Esempi di domande

Riassunto

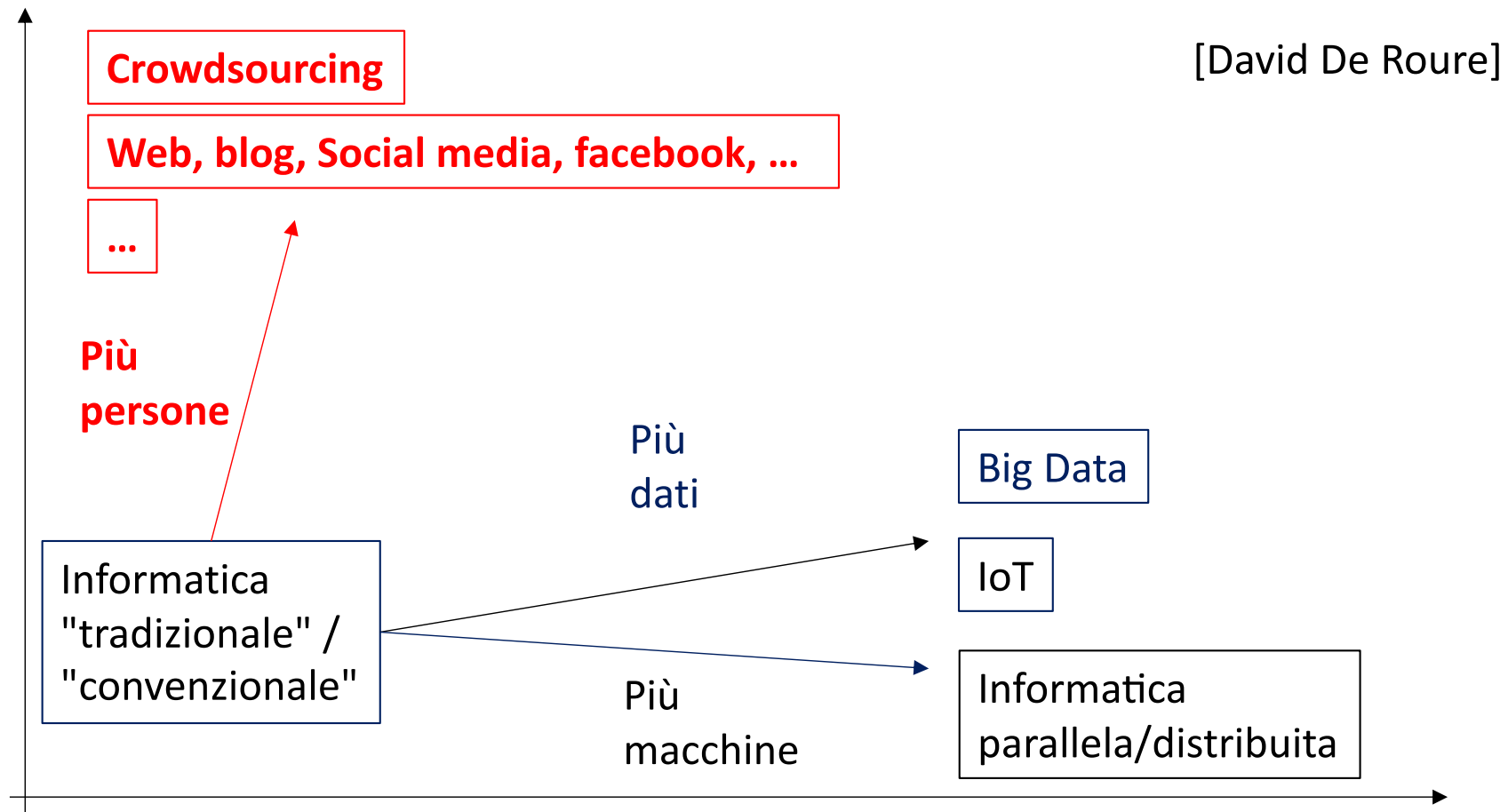
- Corso diviso in due

$$\begin{array}{r} \text{Social Media} \quad + \\ \text{Crowdsourcing} = \\ \hline \text{Social Computing} \end{array}$$

Obiettivi del corso, dalla 1a lezione

- Cosa succede quando i sistemi computazionali vengono a contatto con i sistemi sociali
 - "Folle/masse di persone che usano sistemi computazionali"
 - "Tante persone che usano tanti computer"

Tutto il corso in una slide...



Obiettivi del corso v2, dalla 1a lezione

- Social media (Facebook, Twitter, Instagram, ecc.)
 - Il **comportamento sociale** viene supportato dai **sistemi computazionali**
 - Le persone comunicano sfruttando i sistemi computazionali
 - Parte I
- Crowdsourcing
 - I **sistemi computazionali** vengono supportati dal **comportamento sociale**
 - Compiti che i calcolatori non sanno fare bene vengono svolti da folle di persone raggiunte tramite i sistemi computazionali
 - 🤔
 - Parte II
- Due facce della stessa medaglia

Riassunto – Parte I

- 1. Intro a Social Computing
- 2. Intro a Social Media
 - Termini, Definizioni, Esempi, Classificazioni, Storia, Statistiche
- 3. Grafi e Reti
 - Definizioni di base, Rete = Grande grafo nel mondo reale
- 4, 5, 6. Misure delle reti
 - Centralità, transitività e reciprocità, C, distanza media, degree distribution, ...
- 7, 8, 9. Modelli delle reti
 - Regolari, casuali, piccolo mondo, scale-free/power-law
- 10. Comunità. Il grafo di Facebook
- 11, 12, 13, 14. Diffusione informazioni
 - Herd behavior, Information cascade (ICM), Epidemie & Contagio sociale, Benefici diretti
- 15. Disinformazione, "Fake news".
- 16, 17 Lab. [Hafsa]
 - Python, NetworkX. Pyvis. Progetto
- (Nozioni, ma anche "modo di ragionare")

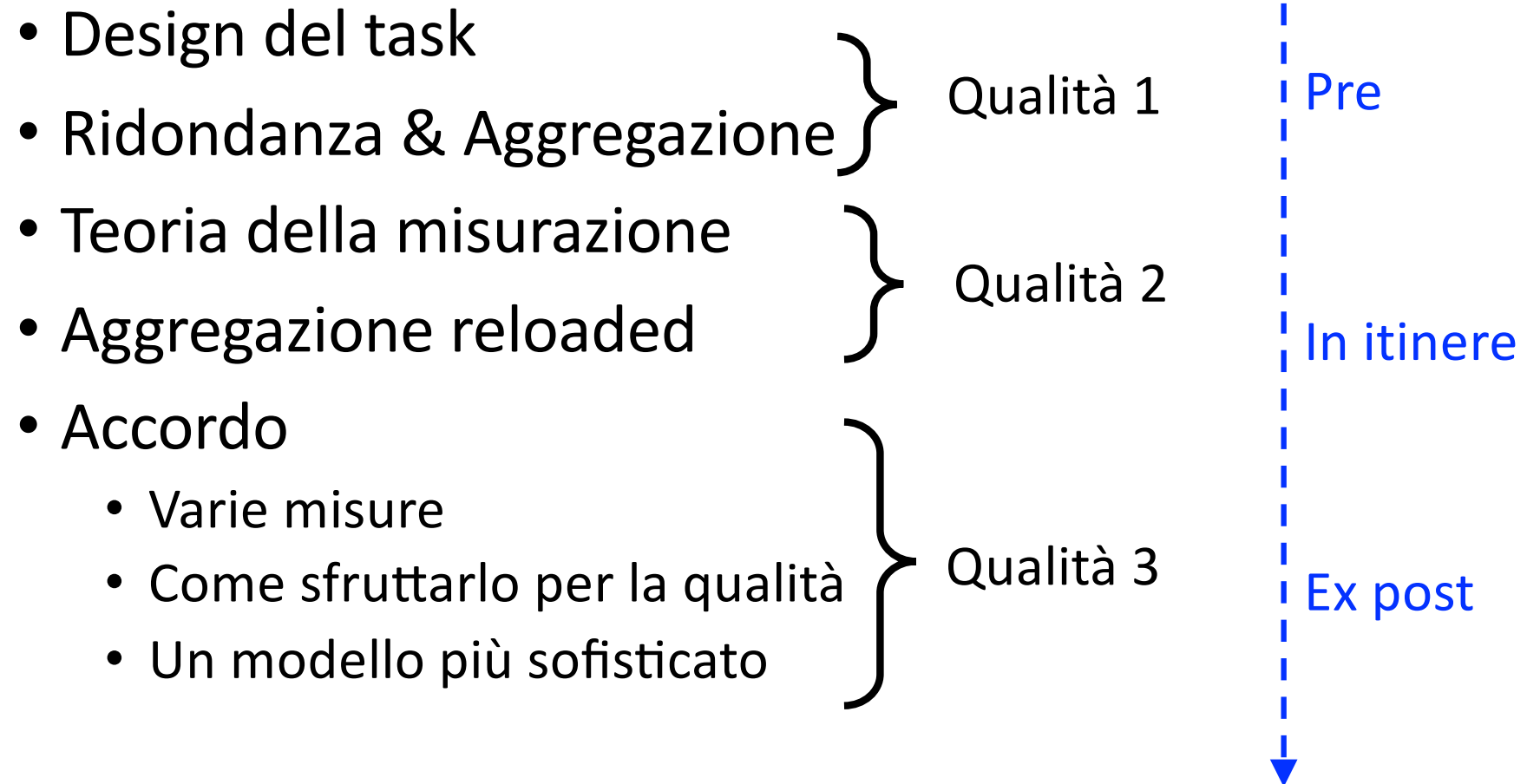
Strumenti e risultati

- Avete un po' di strumenti per
 - ~~Scaricare dati da reti sociali~~
 - Analizzare, modellare, capire, studiare... dati da reti sociali
- Vi ho raccontato (accennato) qualche risultato
 - Grafo di Facebook
 - Disinformazioni
- (Nozioni, ma anche "modo di ragionare")

Riassunto – Parte II

- 18. Intro Crowdsourcing
- 19. Tipologie di Crowdsourcing
- 20. (Micro)task design
- 21. Qualità 1
 - Task design
 - Ridondanza & Aggregazione
- 22. Qualità 2
 - Teoria della misurazione
 - Aggregazione
- 23a. [Michael Soprano] Crowd_Frame
- 23b. Qualità 3
 - Accordo
- 24. Fine
 - Ancora "Fake news"
 - Riassunto
 - Esempi di domande d'esame

Schema



Metodo e strumenti

- Sapete cos'è il crowdsourcing
- Sapete come progettare un esperimento
 - Nozioni metodologiche, qualità
- ~~Sapete~~ Avete qualche idea su come eseguirlo
 - mturk
 - Crowd_Frame
- Sapete un po' come è fatto Crowd_Frame
 - Per usarlo meglio
 - Eventuali argomenti di tesi
- Spero apprezziate lo sforzo (di Michael e ...)
 - Alternativa: fare esperimenti banali, "impazzire" con mturk, ...
 - Avete una possibilità unica
 - Anche per tesi ecc.

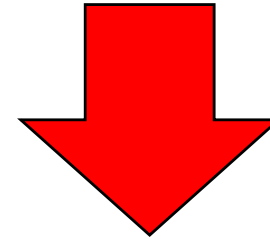
"Lessons learned"

- Varie competenze pratiche
- Avete fatto esperienza
- Non sapete tutto ma sapete abbastanza per cominciare
- Avete messo in pratica alcuni concetti teorici
 - Qualità è importante
 - Qualità perfetta non esiste (in crowdsourcing)
 - Ma "a sufficienza" è sufficiente
 - Non viene gratis
 - "Programmare umani/folle" è
 - complicato
 - ≠ dai computer (ad es., importanza di conoscere lingue, HCI, ...)

Scaletta

- Fine accordo
- Fake news
- Riassunto corso
- **Esame**
 - Regole
 - Esempi di domande

Modalità d'esame



- Scorsi anni:
 - Parte 1 (10pt): attività durante il corso
 - 2 Progettini (5pt + 5pt)
 - Parte 2 (15pt): scritto
 - Ammesso orale se ≥ 8 pt
 - Parte 3 (8pt): orale
 - A discrezione del docente: (+/-5pt)

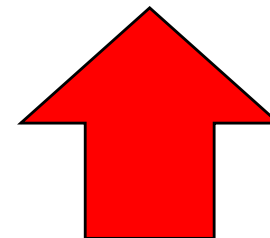
- Quest'anno:
 - Parte 1 (**6**pt): attività durante il corso
 - 1 Progetto
 - Parte 2 (**18**pt): scritto
 - Ammesso orale se ≥ 9 pt
 - Parte 3 (**9**pt): orale
 - A discrezione del docente: (+/-**5**pt)

~20%

~55%

~25%

$\pm$ ~15%



Regole per l'esame

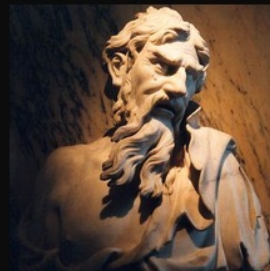
- Progetto
 - Già sapete
 - Valido per tutto l'A.A. e oltre ("Finché sono io il docente")
- Scritto
 - Domande chiuse e aperte
 - Su **tutto** il programma
 - Includere le parti di laboratorio
 - Durata: 2h ca.
- Orale
 - Domande su tutto il programma e sui progetti
- Parti di esame superate restano valide

~~Simulazione d'esame~~

- ~~• Se dovessi fare l'esame oggi come lo farei~~
- Diciamo piuttosto...

Esempi di domande d'esame

- "Ripasso"
- (ma all'esame avrete probabilmente qualcosa di diverso)
- Ad es.:
 - Quasi sicuramente no scelta multipla
 - Probabilmente con risposte più lunghe (1 pag.)



If you do not expect the unexpected, you will not recognize it when it arrives.

~ Heraclitus

AZ QUOTES

Domanda 1

Secondo la definizione di Kietzmann et al. [2011], quali sono le funzionalità del modello ad alveare (honeycomb) dei social media?

1. Presence, Relationship, Reputation, Groups, Conversations, Sharing
2. Presence, Relationship, Reputation, Groups, Conversations, Sharing, Identity
3. Relationship, Reputation, Groups, Conversations, Sharing, Identity
4. Anonymity, Relationship, Reputation, Groups, Conversations, Sharing, Identity

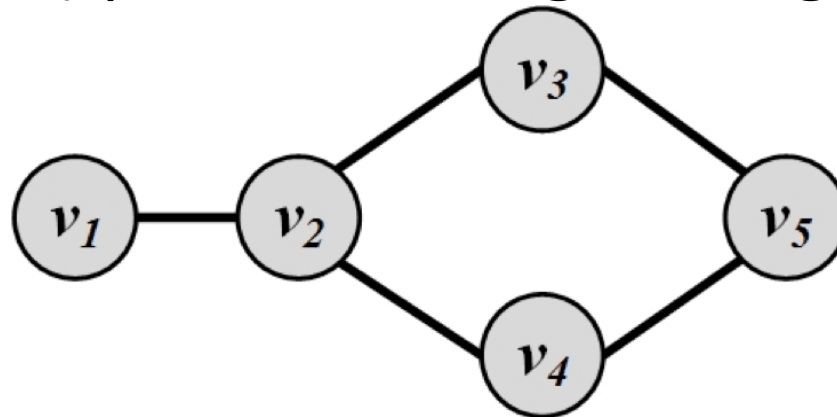
Domanda 2

Quanti sono gli utenti attivi su Facebook secondo le statistiche viste a lezione?

1. Più di 1 miliardo
2. Fra 500 milioni e 1 miliardo
3. Fra 100 milioni e 500 milioni
4. Meno di 100 milioni

Domanda 3

Quali sono i valori di closeness centrality (centralità di vicinanza) per i nodi del grafo seguente?



1. $v1 = 0.5, v2 = 0.8, v3 = 0.66, v4 = 0.66, v5 = 0.57$
2. $v1 = 0.5, v2 = 0.8, v3 = 0.66, v4 = 0.66, v5 = 0.66$
3. $v1 = 0.5, v2 = 0.8, v3 = 0.66, v4 = 0.57, v5 = 0.66$
4. $v1 = 0.5, v2 = 0.8, v3 = 0.57, v4 = 0.57, v5 = 0.57$
5. $v1 = 0.8, v2 = 0.5, v3 = 0.57, v4 = 0.57, v5 = 0.57$

Domanda 4

Qual è la distribuzione dei gradi dei Grafi casuali di Erdos-Renyi?

1. Power-law
2. Poisson (binomiale)
3. Uniforme

Domanda 5

È possibile dichiarare il set $x = \{"1", "1", 2\}$ in Python?

1. Sì
2. No, poiché contiene tipi diversi
3. No, poiché contiene elementi duplicati
4. No, poiché contiene elementi duplicati e tipi diversi

Domanda 6

Usando la libreria NetworkX, a cosa si possono assegnare proprietà in un grafo?

1. Solo ai nodi
2. Solo agli archi
3. A entrambi

Domanda 7

Fra i vari modelli per la diffusione delle informazioni, qual è la differenza principale fra l'herding behavior (comportamento di gregge) e l'information cascade (catene di informazioni?)

1. Herding behavior presuppone una rete diretta, information cascade una rete indiretta
2. Herding behavior si basa su informazioni globali della rete, information cascade su informazioni locali della rete
3. Herding behavior si basa su informazioni locali della rete, information cascade su informazioni globali della rete

Domanda 8

Nel modello di Morris, che relazione c'è fra cluster bloccanti e cascade complete?

1. Non c'è nessuna relazione
2. C'è una cascade completa se e solo se c'è un cluster bloccante
3. C'è una cascade completa se e solo se non c'è un cluster bloccante
4. Se c'è una cascade completa allora c'è un cluster bloccante, ma può esserci un cluster bloccante senza una cascade completa
5. Se c'è un cluster bloccante allora c'è una cascade completa, ma può esserci una cascade completa senza un cluster bloccante

Domanda 9

Quali sono i quattro elementi che rendono una folla saggia secondo Surowiecki?

1. Uniformità di opinioni, Dipendenza, Centralizzazione, Accordo
2. Diversità di opinioni, Indipendenza, Decentralizzazione, Aggregazione
3. Omogeneità, Divisione, Imitazione, Emotività
4. Attenzione, Concentrazione, Ritmo e Vitalità

Domanda 10

Nella teoria della misurazione, quali sono le quattro scale standard?

1. Nominal, Categorical, Interval, Ratio
2. Nominal, Ordinal, Interval, Ratio
3. Nominal, Categorical, Length, Ratio

Domanda 11

Nella piattaforma Amazon Mechanical Turk, chi è un **requester**?

1. Chi progetta il task
2. Chi esegue il task
3. Chi paga i worker affinché eseguano un task progettato

Domanda (aperta) 12

Fra le varie tecniche per il controllo di qualità del microtask crowdsourcing, che cosa sono e a cosa servono i test nascosti? Descriverli e fornire un esempio.

Domanda (aperta) 13

Data la seguente matrice worker / task, qual è l'accordo percentuale (Percent agreement)?
Riportare tutti i passaggi

	t1	t2	t3	t4	t5	t6	t7	t8
w1	Y	N	N	N	N	Y	Y	N
w2	N	N	Y	Y	N	Y	N	N

Domanda (aperta) 14

- Data la seguente matrice worker / task, contenente risposte sulla scala ordinale a tre livelli “Poco”, “Abbastanza” e “Molto”, si eseguano le aggregazioni possibili usando media e moda e se ne discutano pro e contro

	t1	t2	t3	t4	t5
w1	Poco	Molto	Abbastanza	Poco	Poco
w2	Abbastanza	Molto	Abbastanza	Poco	Abbastanza
w3	Poco	Poco	Abbastanza	Molto	Molto

Risposte corrette

1. 2

2. 1

3. 1

4. 2

5. 1

6. 3

7. 2

8. 3

9. 2

10. 2

11. 3

Risposta 12

- Sono domande simili alle altre domande del task, ma la cui risposta è chiara e nota
- Servono a verificare che il worker esegua il compito in modo onesto, senza cercare di imbrogliare
- Se il worker sbaglia i test nascosti – o, in certi casi, ne sbaglia troppi – vuol dire che non sta fornendo risposte corrette, probabilmente perché non intende fare il task in modo onesto ma solo guadagnare il compenso con il minimo sforzo o perché non ha le capacità per eseguire il task o magari perché è un bot
- Esempio: in un task in cui si chiede di verificare la veridicità di alcune affermazioni, chiedere di farlo per un'affermazione palesemente vera o falsa

Risposta 13

	t1	t2	t3	t4	t5	t6	t7	t8
w1	Y	N	N	N	N	Y	Y	N
w2	N	N	Y	Y	N	Y	N	N

- 1) Costruire la confusion matrix

		w2	
		Y	N
w1	Y	1	2
	N	2	3

- 2) Percent agreement = $(1+3)/(1+2+2+3)=4/8=50\%$

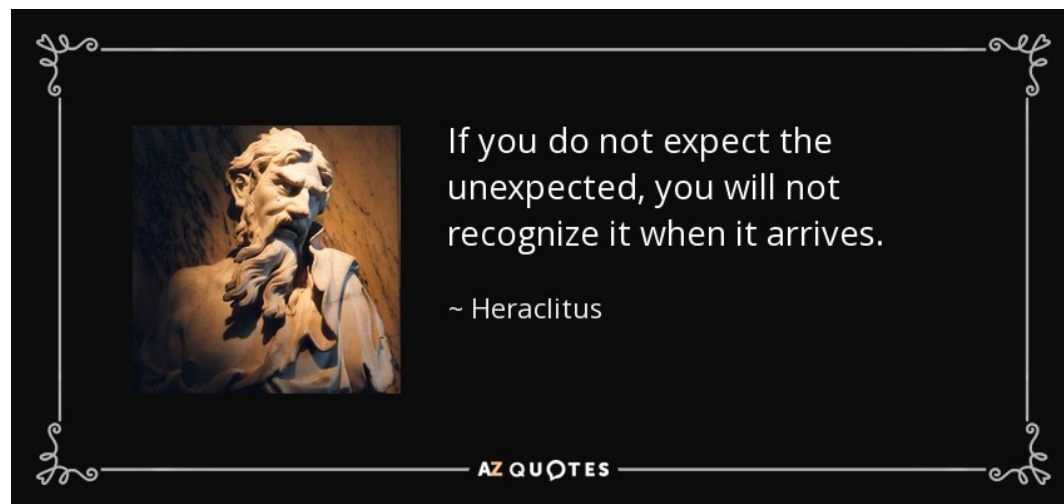
Risposta 14

	t1	t2	t3	t4	t5
w1	0	2	1	0	0
w2	1	2	1	0	1
w3	0	0	1	2	2

- Per aggregare con medie devo trasformare "Poco", "Abbastanza" e "Molto" in valori numerici (scelta arbitraria: 0, 1, 2)
 - Media: $1/3, 4/3, 1, 2/3, 1$
 - Moda: 0, 2, 1, 0, 1 (arbitrario)
- Pro media: si riesce a differenziare i casi t1 vs. t4; non si ha arbitrarietà per t5.
- Contro media: la scala è ordinale e quindi calcolare la media è scorretto in generale perché assume categorie / livelli equidistanti; produce valori che non appartengono alla scala
- Pro Moda: è un'operazione corretta per la scala ordinale; produce valori aggregati che appartengono alla scala
- Contro Moda: Perde informazioni e ad esempio non è in grado di differenziare i casi t1 e t4

Commenti

- 11 domande chiuse
- 3 domande aperte
 - CIRCA 1 punto / domanda, circa 1' / domanda
 - (Possibili penalizzazioni per risposte strampalate)
- Ma sono possibili / probabili anche domande ben diverse (e magari più difficili!)
 - Vi do la definizione di una **nuova** misura di centralità e vi chiedo di calcolarla su una rete
 - Vi faccio leggere un articolo / brano e vi chiedo qualcosa di specifico



N.B. Le domande coprono **tutti** gli argomenti del corso

- Social Media Intro [Lezione 2]
- Misure sulle reti [3-6]
- Modelli Reti [7-9]
- Grafo Facebook [10]
- Diffusione (dis)informazioni [11-15]
- Python, NetworkX, Pyvis [16, 17]
- Crowdsourcing in generale [18-20]
- Qualità e crowdsourcing, Measurement Theory, Aggregazione, Accordo [21-24]
- mturk, Crowd_Frame [23]

Esempio d'esame

- (vedi pdf su teams)
- Ma voto in /30, invece che in /18
- Ma quest'anno abbiamo visto meno crowdsourcing degli anni scorsi, quindi probabilmente ce ne sarà di meno
 - Da 50% – 50% a 66% – 33% (circa!)

Scaletta

- Fine accordo
- Fake news
- Riassunto corso
- Esame
 - Regole
 - Esempi di domande

Fine

- 🙋 Studiate! 📖📚🧐🧐🧐